

PERMUTATION RESOLUTIONS AND COHOMOLOGICAL SINGULARITY

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ABSTRACT. We consider the bounded derived category of finitely generated representations of a finite group, with coefficients in a noetherian commutative ring. When the coefficient ring in question is regular we prove that this derived category can be built from a simpler one, namely the homotopy category of *permutation* modules, via localization and idempotent-completion. When the coefficient ring is singular, we construct an invariant to decide when a complex reduces to permutation modules. This invariant uses group cohomology and takes values in the big singularity category of the coefficient ring.

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1. INTRODUCTION

In the whole paper, G is a finite group and R is a commutative noetherian ring.

In this generality, the group algebra RG combines the complexity of commutative algebra and of representation theory. Even ignoring the G -action, one does not expect a simple description of finitely generated R -modules. And even when we have one, say when R is a field, if that field has characteristic $p > 0$ dividing the order of G , we still face the wild complexity of modular representation theory.

Much tidier is the class of *permutation RG -modules*, i.e. those isomorphic to a free R -module $R(A)$, with basis a finite G -set A and with G -action R -linearly extended from the one on A . Any such module is isomorphic to $R(G/H_1) \oplus \cdots \oplus R(G/H_r)$ for a choice of $H_1, \dots, H_r$ among the finitely many subgroups of G .

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Despite their apparent simplicity, permutation modules play a non-trivial role in many areas of group representation theory, as recalled for instance in the introduction of Benson-Carlson [BC20]. Our own interest in permutation modules stems from their connection with Artin motives, following Voevodsky [Voe00, §3.4].

Let us consider the inclusion of the additive category of permutation RG -modules into the abelian category of finitely generated RG -modules, denoted as follows:

$$(1.1) \quad \text{perm}(G; R) \subseteq \text{mod}(RG)$$

It gives a canonical functor $F: K_b(\text{perm}(G; R)) \rightarrow D_b(\text{mod}(RG))$ from the bounded homotopy category of the former to the bounded derived category of the latter. The kernel of F is the thick subcategory $K_{b,ac}(\text{perm}(G; R))$ of *acyclic* complexes of permutation modules, studied in [BC20]. The functor F descends to the corresponding Verdier quotient and, after idempotent-completion $(\dots)^{\natural}$, yields a functor

$$(1.2) \quad \bar{F}: \left(\frac{K_b(\text{perm}(G; R))}{K_{b,ac}(\text{perm}(G; R))} \right)^{\natural} \longrightarrow D_b(\text{mod}(RG)).$$

This canonical functor $\bar{F}$ is our main object of study. The only formal property that $\bar{F}$ inherits by construction is being conservative. So the first surprise is:

1.3. Theorem. *The canonical functor $\bar{F}$ of (1.2) is always fully faithful.*

This is part of our main result that furthermore identifies the image of $\bar{F}$. For an object X in $D_b(\text{mod}(RG))$ to be in the image of $\bar{F}$, it is clearly necessary for its underlying complex of R -modules $\text{Res}_1^G(X)$ to be *perfect* over R . For $G = 1$, this is obviously the only condition. However we prove in Example 5.16 that this naive necessary condition is not sufficient for general groups. To state the correct result, we invoke the *singularity categories* of the ring R (which do not depend on G)

$$D_b^{\text{sing}}(R) := \frac{D_b(\text{mod}(R))}{D_{\text{perf}}(R)} \quad \text{and} \quad D^{\text{sing}}(R) := K_{ac}(\text{Inj}(R)).$$

The construction of the right-hand ‘*big*’ *singularity category* $D^{\text{sing}}(R)$, following Krause [Kra05], can be found in Recollection 4.1. It is a compactly generated triangulated category, whose subcategory of compact objects coincides with the idempotent-completion of the left-hand $D_b^{\text{sing}}(R)$, the perhaps more familiar (*small*) *singularity category*; see Orlov [Orl04]. The name ‘singularity category’ comes from the fact that R is regular if and only if $D^{\text{sing}}(R) = D_b^{\text{sing}}(R) = 0$. More generally, Stevenson [Ste14] relates $D_b^{\text{sing}}(R)$ to the singularity locus of R via tensor-triangular geometry. Krause also extends the obvious quotient functor $D_b(\text{mod}(R)) \rightarrow D_b^{\text{sing}}(R)$ to unbounded complexes. We call this extension the *singularity functor*

$$\text{sing}: D(R) \rightarrow D^{\text{sing}}(R).$$

For each subgroup $H \leq G$, let $(-)^{hH}: D(RG) \rightarrow D(R)$ be the right-derived functor of the H -fixed-points functor $(-)^H$. We can now state our main result.

1.4. Theorem (Theorem 6.16). *The canonical functor $\bar{F}$ of (1.2) is fully-faithful and its essential image consists of those $X \in D_b(\text{mod}(RG))$ such that the invariant*

$$(1.5) \quad \chi^H(X) := \text{sing}(X^{hH})$$

vanishes in the big singularity category $D^{\text{sing}}(R)$, for every subgroup $H \leq G$.

The functor $(-)^{hG} : D(RG) \rightarrow D(R)$ is the derived-category version of ordinary group cohomology, that is, the following left-hand square commutes:

$$(1.6) \quad \begin{array}{ccc} \text{Mod}(RG) & \hookrightarrow & D(RG) \\ \downarrow H^*(G, -) & & \downarrow (-)^{hG} \\ \text{Mod}(R) & \xleftarrow{H^*} & D(R) \end{array} \quad \begin{array}{c} \searrow \chi^G \\ \text{(def)} \end{array} \quad \begin{array}{c} \xrightarrow{\text{sing}} \\ D^{\text{sing}}(R) \end{array}$$

We call the invariant $\chi^H : D(RG) \xrightarrow{\text{Res}_H^G} D(RH) \xrightarrow{(-)^{hH}} D(R) \xrightarrow{\text{sing}} D^{\text{sing}}(R)$ of (1.5) the *H-cohomological singularity* functor, for every subgroup $H \leq G$. To apply Theorem 1.4 to an X whose underlying complex $\text{Res}_1^G(X)$ is perfect it suffices to test $\chi^H(X) = 0$ for the Sylow subgroups H of G , or alternatively for the (maximal) elementary abelian subgroups. In particular, if G is a p -group, there are two conditions for a complex X to belong to the image of $\bar{F}$: the naive $\text{Res}_1^G(X) \in D_{\text{perf}}(R)$ and the new $\chi^G(X) = 0$ in $D^{\text{sing}}(R)$.

To appreciate the strength of Theorem 1.4, observe that when the ring R is regular, the condition $\chi^H(X) = 0$ is trivially true in $D^{\text{sing}}(R) = 0$. And yet this recovers a non-trivial result due to Rouquier (unpublished; see [BV08, § 2.4] or <https://www.math.ucla.edu/~rouquier/papers/perm.pdf>).

1.7. Corollary (Rouquier; see Corollary 6.20). *Let R be a regular commutative ring. The canonical functor $\bar{F}$ of (1.2) is a triangulated equivalence*

$$\left(\frac{K_b(\text{perm}(G; R))}{K_{b,ac}(\text{perm}(G; R))} \right)^{\natural} \xrightarrow{\sim} D_b(\text{mod}(RG)).$$

In fact, we give here two proofs of this fact: one as a corollary of Theorem 1.4 and a more direct one which does not use cohomological singularity, in Scholium 3.15.

In the even more special case where $R = k$ is a field, the main result of Balmer-Benson [BB20] says that every kG -module M admits a ‘complement’ N such that $M \oplus N$ possesses a finite resolution by permutation modules. We give a derived-category reformulation of this fact in Corollary 3.11, that implies Corollary 1.7 for $R = k$. Corollary 3.11 also yields some control on the complement N of [BB20] in terms of M , using Thomason’s classification [Tho97, § 2] of ‘dense’ triangulated subcategories via Grothendieck groups. As application, we obtain for instance Corollary 8.7, whose statement does not involve derived categories:

1.8. Corollary. *Let M be a finitely generated kG -module, where k is a field. Consider the Heller loop of M defined by $\Omega(M) := \text{Ker}(kG \otimes_k M \rightarrow M)$. Then $M \oplus \Omega(M)$ admits a finite resolution $0 \rightarrow P_n \rightarrow \cdots \rightarrow P_0 \rightarrow M \oplus \Omega(M) \rightarrow 0$ where all P_i are finitely generated permutation kG -modules.*

Replacing permutation by p -permutation modules (a. k. a. trivial source modules) we will also prove the following striking result (Theorem 8.17).

1.9. Theorem. *Let k be a field of characteristic $p > 0$. Every finitely generated kG -module admits a finite p -permutation resolution.*

In particular for p -groups the ‘complement’ N of [BB20] can be taken to be zero:

1.10. Corollary. *If the field k has characteristic $p > 0$ and G is a p -group then every finitely generated kG -module has a finite resolution by permutation modules.*

Finally, let us mention a variant for ‘big’ triangulated categories. The Neeman-Thomason Theorem [Nee92] suggests that the equivalence of Corollary 1.7 is the compact tip of an iceberg involving compactly generated triangulated categories. Indeed, Krause [Kra05] proved that $D_b(\text{mod}(RG))$ is the category of compact objects in the homotopy category of injectives $K(\text{Inj}(RG))$. So we expect the existence of a ‘big’ triangulated category $D\text{Perm}(G; R)$ whose compact objects would consist of $K_b(\text{perm}(G; R))^{\natural}$ and such that the finite localization with respect to $K_{b,ac}(\text{perm}(G; R))$ coincides with $K(\text{Inj}(RG))$. This is what we prove in Corollary 7.17 for R regular. We construct $D\text{Perm}(G; R)$ and describe the corresponding finite localization even when R is singular. A more thorough exposition will appear in the companion paper [BG20], where we also explain the connection with cohomological Mackey functors and with Artin motives in some detail.

The organization of the paper is as follows. We discuss a refinement of the category of permutation modules in Section 2, that is technically more convenient. We identify it with the essential image of $\bar{F}$ in Section 3. Section 4 is dedicated to Krause’s big singularity category and Section 5 discusses our ‘cohomological singularity’ functor (1.5). We prove our main result in Section 6 and the variant for big categories in Section 7. Finally we deduce the corollaries about permutation resolutions in Section 8. Note that we already prove the field case of our main result in Section 3. So the field-focussed reader can follow the shortcut Section 2 $\rightarrow$ Section 3 $\rightarrow$ Section 8.

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Notation and convention.

We write $\simeq$ for isomorphisms and reserve $\cong$ for canonical isomorphisms.

A commutative noetherian ring R is *regular* if it is locally of finite projective dimension. Most results reduce to the case where R is connected (not a product).

Unless specified, modules are left modules. We denote by $\text{Mod}(?)$ the category of modules and by $\text{mod}(?)$ the subcategory of finitely generated ones.

Since fixed points $(-)^H$ and other decorations (duals) appear in exponent, we use homological notation for complexes $\cdots \rightarrow M_n \rightarrow M_{n-1} \rightarrow \cdots$. We write K (and K_b) for homotopy categories of (bounded) complexes, and D (and D_b) for (bounded) derived categories. We abbreviate $D_b(RG)$ for $D_b(\text{mod}(RG))$. When we speak of a module M as a complex, we mean it concentrated in degree zero.

All triangulated subcategories are implicitly assumed to be replete. We abbreviate ‘thick’ for ‘triangulated and thick’ (i.e. closed under direct summands). A triangulated subcategory is called localizing if it is closed under coproducts.

We denote by $\mathcal{A}^{\natural}$ the idempotent-completion (Karoubi envelope) of an additive category $\mathcal{A}$, or its obvious realization in some ambient idempotent-complete category. See [BS01] for details, including the fact that $K_b(\mathcal{A}^{\natural}) \cong K_b(\mathcal{A})^{\natural}$.

1.11. *Recollection.* We defined the category $\text{perm}(G; R)$ of permutation RG -modules before (1.1). If G is a p -group and $R = k$ is a field of characteristic p , this category is idempotent-complete ([Fei82, IX, 3.4]). For a general group G , summands of permutation modules are called *p-permutation modules*, or *trivial source modules*. A kG -module is *p-permutation* if and only if its restriction to a p -Sylow subgroup is

a permutation module. This characterization is specific to fields of characteristic p whereas the idempotent-completion $\text{perm}(G; R)^{\natural}$ makes sense for any ring R .

We use the phrase ‘ M is $\natural$ -permutation’ to say that M belongs to $\text{perm}(G; R)^{\natural}$.

2. BOUNDED PERMUTATION RESOLUTIONS

Until specified otherwise, *we assume all RG -modules finitely generated and all complexes bounded*. As in [BB20], we begin with a stronger notion of resolution.

2.1. Definition. Let X be a bounded complex of RG -modules and $m \in \mathbb{Z}$. An m -free permutation resolution of X is a quasi-isomorphism of complexes $s: P \rightarrow X$ where P is a bounded complex of permutation RG -modules which is m -free, meaning that P_i is free for all $i \leq m$. Clearly m' -free implies m -free when $m' \geq m$.

Similarly (Recollection 1.11) an m -projective $\natural$ -permutation resolution is a quasi-isomorphism $P \rightarrow X$ where all P_i are $\natural$ -permutation, and projective for $i \leq m$.

2.2. Remark. The statements of Lemma 2.4, Corollary 2.5 and Proposition 2.7 below also hold with the words ‘permutation’ replaced by ‘ $\natural$ -permutation’ and with ‘free’ replaced by ‘projective’. We leave most of their proofs to the reader.

2.3. Remark. The word ‘resolution’ in Definition 2.1 can be misleading for the complex P is allowed to extend further to the right than X itself, even for $X = M[0]$, a single RG -module in degree zero. This can be corrected, when m is large enough:

2.4. Lemma. *Let $m \geq n$ be such that the complex X is acyclic strictly below degree n and such that X admits an m -free permutation resolution. Then X admits an m -free resolution $P \rightarrow X$ where $P_i = 0$ for all $i < n$. (See Remark 2.2.)*

Proof. We can assume $n = 0$. So $m \geq 0$. Let $s: Q \rightarrow X$ be an m -free permutation (resp. m -projective $\natural$ -permutation) resolution. Since s is a quasi-isomorphism, $H_i(Q) = 0$ for $i < 0$. Since Q_i is projective for $i < 0$, the complex Q ‘splits’ in negative degrees. So we have a decomposition $Q = Q' \oplus Q''$ where the subcomplex $Q' = \cdots \rightarrow Q_1 \rightarrow Q'_0 \rightarrow 0 \rightarrow \cdots$ is concentrated in non-negative degrees whereas $Q'' = \cdots \rightarrow 0 \rightarrow Q''_0 \rightarrow Q_{-1} \rightarrow \cdots$ lives in non-positive degrees and is acyclic, i.e. the composite $Q' \rightarrow Q \rightarrow X$ remains a quasi-isomorphism. At this stage Q'_0 is only projective but not necessarily free. (In the case of m -projective $\natural$ -permutation resolutions, the proof stops here with $P = Q'$.) Since Q'' is split exact, we see that Q''_0 is stably free: $Q''_0 \oplus (\oplus_{i < 0, \text{even}} Q_i) \simeq \oplus_{i < 0, \text{odd}} Q_i$. Since $Q'_0 \oplus Q''_0 = Q_0$ is free, we see that Q'_0 is also stably free, namely $Q'_0 \oplus L$ is free for the free $L = \oplus_{i < 0, \text{odd}} Q_i$. Adding to Q' the complex $0 \rightarrow L \xrightarrow{1} L \rightarrow 0$ with L in degrees 1 and 0 (with zero map to X), we obtain a new complex P and a quasi-isomorphism $P \rightarrow X$, where now $P_0 = Q'_0 \oplus L$ is free and $P_1 = Q_1 \oplus L$ is permutation (and free if Q_1 was) and $P_i = Q_i$ for $i > 1$ and $P_i = 0$ for $i < 0$. This $P \rightarrow X$ is the desired resolution. $\square$

2.5. Corollary. *Let $M \in \text{mod}(RG)$ be such that, when viewed as a complex, M admits a 0-free permutation resolution in the sense of Definition 2.1. Then M admits a finite resolution by finitely generated permutation modules. (See Remark 2.2.)* $\square$

2.6. Lemma. *Let P, X, Y be bounded complexes, let $f: P \rightarrow X$ be a morphism and let $s: Y \rightarrow X$ be a quasi-isomorphism:*

$$\begin{array}{ccc} & \hat{f} & Y \\ P & \xrightarrow{\quad} & \downarrow s \\ & f & X \end{array}$$

Suppose that $m \in \mathbb{Z}$ is such that $X_i = 0$ and $Y_i = 0$ for $i > m$ and P_i is projective for all $i \leq m$. Then there exists $\hat{f}: P \rightarrow Y$ such that $s\hat{f}$ is homotopic to f .

Proof. Let Z be an acyclic complex such that $Z_i = 0$ for $i > m + 1$. Then any map $P \rightarrow Z$ is null-homotopic, as one can build a homotopy using the usual induction argument, that only requires P_i projective for $i \leq m$ to lift against the epimorphism $Z_{i+1} \twoheadrightarrow \text{im}(Z_{i+1} \rightarrow Z_i)$. Now $Z := \text{cone}(s)$ is such a complex. So the composite $P \xrightarrow{f} X \rightarrow \text{cone}(s)$ is zero in $K_b(\text{mod}(RG))$. Hence f factors as claimed. $\square$

2.7. Proposition. *Let $s: Y \rightarrow X$ be a quasi-isomorphism. Then X admits m -free permutation resolutions for all $m \geq 0$ if and only if Y does. (See Remark 2.2.)*

Proof. From Y to X is trivial. So suppose that X has the property and let us show it for Y . Let $m \geq 0$. Increasing m if necessary, we can assume that $X_i = Y_i = 0$ for all $i > m$. Let then $f: P \rightarrow X$ be an m -free permutation resolution of X . By Lemma 2.6, there exists $\hat{f}: P \rightarrow Y$ such that $s\hat{f} \sim f$. By 2-out-of-3, $\hat{f}: P \rightarrow Y$ is a quasi-isomorphism, hence yields an m -free permutation resolution of Y . $\square$

In Definition 2.1, we dealt with complexes on the nose. The above proposition allows us to pass to the derived category, if we make sure to require the existence of m -free permutation resolutions for *all* $m \geq 0$, not just for some specific m .

2.8. Definition. We have well-defined replete subcategories of the derived category

$$\begin{aligned} \mathcal{P}(G; R) &= \left\{ X \in D_b(RG) \left| \begin{array}{l} X \text{ admits } m\text{-free permutation resolutions} \\ \text{in the sense of Definition 2.1, for all } m \geq 0 \end{array} \right. \right\}, \\ \mathcal{Q}(G; R) &= \left\{ X \in D_b(RG) \left| \begin{array}{l} X \text{ admits } m\text{-projective } \mathfrak{k}\text{-permutation resolutions} \\ \text{in the sense of Definition 2.1, for all } m \geq 0 \end{array} \right. \right\}. \end{aligned}$$

2.9. Proposition. *The two subcategories $\mathcal{P}(G; R) \subseteq \mathcal{Q}(G; R)$ above are triangulated subcategories of $D_b(RG)$.*

Proof. We prove it for $\mathcal{P}(G; R)$. The proof for $\mathcal{Q}(G; R)$ is similar. It suffices to show that if $f: X \rightarrow Y$ is a morphism in $D_b(RG)$ with $X, Y \in \mathcal{P}(G; R)$ then $\text{cone}(f) \in \mathcal{P}(G; R)$. The morphism f is represented by a fraction $X \xleftarrow{s} Z \xrightarrow{g} Y$ in $\text{Ch}_b(RG)$ with s a quasi-isomorphism. By Proposition 2.7, we have $Z \in \mathcal{P}(G; R)$. Since $\text{cone}(f) \simeq \text{cone}(g)$ in $D_b(RG)$, we can assume that $f: X \rightarrow Y$ is a plain morphism of complexes. Let now $n \geq 0$. Since Y belongs to $\mathcal{P}(G; R)$, choose an n -free permutation resolution $t: Q \rightarrow Y$. Choose $m \gg n$ such that $Q_i = Y_i = 0$ for $i > m$, using that Q and Y are bounded. Since $X \in \mathcal{P}(G; R)$, choose an m -free permutation resolution $s: P \rightarrow X$. We thus have the (plain) morphisms f, s, t :

$$\begin{array}{ccc} P & \xrightarrow{\quad h \quad} & Q \\ s \downarrow & & \downarrow t \\ X & \xrightarrow{\quad f \quad} & Y \end{array}$$

By Lemma 2.6, there exists $h: P \rightarrow Q$ such that $th \sim fs$. Since s and t are quasi-isomorphisms, so is the induced map $\text{cone}(h) \rightarrow \text{cone}(f)$. Now, the mapping cone of h is a complex of permutation modules that is free in degrees $\leq n$ since P and Q are. As $n \geq 0$ was arbitrary, we proved $\text{cone}(f) \in \mathcal{P}(G; R)$ as claimed. $\square$

2.10. Example. Let G be a p -group and $R = k$ a field of characteristic p . Then $D_b(kG)$ is generated as a triangulated subcategory by k . So, by Proposition 2.9,

the triangulated subcategory $\mathcal{P}(G; k) = \mathcal{Q}(G; k)$ is equal to $D_b(kG)$ if and only if it contains k . We will prove that this holds in Theorem 8.17.

2.11. Recollection. A triangulated subcategory $\mathcal{A} \subseteq \mathcal{T}$ is *dense* if every object X of $\mathcal{T}$ is a direct summand of an object $X \oplus Y$ of $\mathcal{A}$. This amounts to $X \oplus \Sigma X \in \mathcal{A}$ since $X \oplus \Sigma X = \text{cone} \left(\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} : X \oplus Y \rightarrow X \oplus Y \right)$. Hence the thick closure of $\mathcal{A}$ is

$$\text{thick}(\mathcal{A}) = \{ X \in \mathcal{T} \mid \exists Y \in \mathcal{T} \text{ s.t. } X \oplus Y \in \mathcal{A} \} = \{ X \in \mathcal{T} \mid X \oplus \Sigma X \in \mathcal{A} \}.$$

When $\mathcal{T}$ is idempotent-complete (like here $D_b(RG)$) we have $\text{thick}(\mathcal{A}) = \mathcal{A}^\natural$.

2.12. Proposition. *The triangulated subcategory $\mathcal{P}(G; R)$ is dense in $\mathcal{Q}(G; R)$.*

Proof. It suffices to show that for every $X \in \mathcal{Q}(G; R)$, we have $X \oplus \Sigma X \in \mathcal{P}(G; R)$. By Definition 2.1, it suffices to show that if P is an m -projective complex of $\natural$ -permutation RG -modules for some $m \geq 0$ then $P \oplus \Sigma P$ is homotopy equivalent to an m -free complex $\tilde{P}$ of permutation RG -modules. For each i , since P_i is $\natural$ -permutation (resp. projective for $i \leq m$), there exists Q_i such that $P_i \oplus Q_i$ is permutation (resp. free for $i \leq m$). Adding to $P \oplus \Sigma P$ short complexes $\cdots 0 \rightarrow Q_i \xrightarrow{1} Q_i \rightarrow 0 \cdots$, with Q_i in degrees $i+1$ and i , yields the wanted m -free permutation complex $\tilde{P}$. $\square$

2.13. Remark. The thick closure of the subcategory $\mathcal{P}(G; R)$ of $D_b(RG)$

$$\mathcal{P}(G; R)^\natural = \mathcal{Q}(G; R)^\natural = \text{thick}(\mathcal{P}(G; R)) = \text{thick}(\mathcal{Q}(G; R))$$

is a central object in this paper, as it will turn out (Theorem 3.21) to be the essential image of $\bar{F}$ in (1.2). We also want to decide when $\mathcal{P}(G; R)^\natural$ coincides with $D_b(RG)$. More generally, we want to construct an invariant on $D_b(RG)$ that detects $\mathcal{P}(G; R)^\natural$. We sow this invariant in Section 5 and reap the result in Theorem 6.16.

2.14. Example. For $G = 1$, this gives $\mathcal{P}(1; R)^\natural = D_{\text{perf}}(R)$, the perfect complexes.

Let us start with generalities about the Mackey 2-functor $\mathcal{P}(?: R)$.

2.15. Recollection. Recall that for a subgroup $H \leq G$, induction is a two-sided adjoint to restriction. In particular, these functors preserve injectives and projectives and yield well-defined functors on derived categories without need to derive them on either side. Recall also that the composite (for Y a module or a complex over H)

$$(2.16) \quad \begin{array}{ccc} Y & \xrightarrow{\eta^\ell} & \text{Res}_H^G \text{Ind}_H^G(Y) \xrightarrow{\epsilon^r} Y \\ & \searrow \text{id} \nearrow & \end{array}$$

of the (usual) unit for the $\text{Ind} \dashv \text{Res}$ adjunction and the (usual) counit for the $\text{Res} \dashv \text{Ind}$ adjunction is the identity. The other composite (now for X over G)

$$(2.17) \quad \begin{array}{ccc} X & \xrightarrow{\eta^r} & \text{Ind}_H^G \text{Res}_H^G(X) \xrightarrow{\epsilon^\ell} X \\ & \searrow [G:H] \nearrow & \end{array}$$

of the (usual) $\text{Res} \dashv \text{Ind}$ unit and $\text{Ind} \dashv \text{Res}$ counit is multiplication by $[G : H]$.

2.18. Proposition. *Let $H \leq G$ be a subgroup.*

- (a) *For $X \in D_b(RG)$, if $X \in \mathcal{P}(G; R)^\natural$ then $\text{Res}_H^G(X) \in \mathcal{P}(H; R)^\natural$.*
- (b) *For $Y \in D_b(RH)$, we have $Y \in \mathcal{P}(H; R)^\natural$ if and only if $\text{Ind}_H^G(Y) \in \mathcal{P}(G; R)^\natural$.*
- (c) *If $X \in D_b(RG)$ is such that $\text{Res}_H^G(X) \in \mathcal{P}(H; R)^\natural$ and multiplication by $[G : H]$ is invertible on X (e.g. if $[G : H]$ is invertible in R) then $X \in \mathcal{P}(G; R)^\natural$.*

Proof. Direct from Res_H^G and Ind_H^G being exact and preserving permutation modules and free modules, using (2.16) in the ‘if’ part of (b) and using (2.17) in (c). $\square$

2.19. *Definition.* We say that a complex $X \in \text{D}_b(RG)$ is *R-perfect* if the underlying complex $\text{Res}_1^G(X)$ is perfect over R . This defines a thick subcategory of $\text{D}_b(RG)$

$$\text{D}_{R\text{-perf}}(RG) := \{ X \in \text{D}_b(RG) \mid \text{Res}_1^G(X) \in \text{D}_{\text{perf}}(R) \}.$$

2.20. **Corollary.** *We have $\mathcal{P}(G; R)^\natural \subseteq \text{D}_{R\text{-perf}}(RG)$. If the order $|G|$ is invertible in R , then $\mathcal{P}(G; R)^\natural = \text{D}_{R\text{-perf}}(RG)$.*

Proof. Immediate from Proposition 2.18 for $H = 1$ and from Example 2.14. $\square$

Let us say a few words of the tensor product and its preservation of $\mathcal{P}(G; R)$. For the end of the section, we allow arbitrary modules and unbounded complexes.

2.21. *Recollection.* We can ‘tensor over R and use diagonal G -action’ as usual:

$$- \otimes_R -: \text{Mod}(RG) \times \text{Mod}(RG) \longrightarrow \text{Mod}(RG \otimes_R RG) \longrightarrow \text{Mod}(RG).$$

This right-exact tensor can be left-derived (we only use the following generality):

$$- \otimes_R^{\mathbb{L}} -: \text{D}_+(RG) \times \text{D}_+(RG) \longrightarrow \text{D}_+(RG).$$

Given right-bounded complexes $X \in \text{Ch}_+(RG)$ and $Y \in \text{Ch}_+(RG)$, choose projective resolutions $P_X \rightarrow X$ and $P_Y \rightarrow Y$ and define in $\text{D}_+(RG)$:

$$X \otimes_R^{\mathbb{L}} Y := P_X \otimes_R P_Y \cong X \otimes_R P_Y \cong P_X \otimes_R Y.$$

The latter two isomorphisms use that P_X and P_Y are degreewise R -flat. In fact, if either X or $Y \in \text{Ch}_+(RG)$ is degreewise R -flat, we have $X \otimes_R^{\mathbb{L}} Y \cong X \otimes_R Y$.

For a subgroup $H \leq G$, the properties of restriction (Recollection 2.15) give us

$$\text{Res}_H^G(X \otimes_R^{\mathbb{L}} Y) \cong \text{Res}_H^G(X) \otimes_R^{\mathbb{L}} \text{Res}_H^G(Y).$$

2.22. **Proposition.** *Let $P, Q \in \text{Ch}_b(\text{perm}(G; R))$. If P is m -free and Q is n -free then $P \otimes_R Q$ remains in $\text{Ch}_b(\text{perm}(G; R))$ and is $(m + n + 1)$ -free.*

Proof. The tensor of permutation modules is permutation by Mackey and the tensor of a permutation with a free is free by Frobenius. Hence if $\ell \leq m + n + 1$ and $\ell = i + j$ then $i \leq m$ or $j \leq n$ and the summand $P_i \otimes_R Q_j$ of $(P \otimes_R Q)_\ell$ is free. $\square$

2.23. **Corollary.** *The tensor product $\otimes_R^{\mathbb{L}}$ on $\text{D}_+(RG)$ restricts to a functor*

$$- \otimes_R^{\mathbb{L}} -: \mathcal{P}(G; R) \times \mathcal{P}(G; R) \longrightarrow \mathcal{P}(G; R).$$

And similarly $\mathcal{P}(G; R)^\natural \otimes_R^{\mathbb{L}} \mathcal{P}(G; R)^\natural \subseteq \mathcal{P}(G; R)^\natural$.

Proof. If $s: P \rightarrow X$ and $t: Q \rightarrow Y$ are m -free permutation resolutions for $m \geq 0$ then so is $s \otimes_R^{\mathbb{L}} t: P \otimes_R Q \cong P \otimes_R^{\mathbb{L}} Q \xrightarrow{s \otimes^{\mathbb{L}} t} X \otimes_R^{\mathbb{L}} Y$ by Proposition 2.22. $\square$

Here is another use of the tensor, perhaps even more straightforward:

2.24. **Proposition.** *Let $R \rightarrow R'$ be a ring homomorphism. Then the extension-of-scalars functor $R' \otimes_R^{\mathbb{L}} -: \text{D}_+(RG) \rightarrow \text{D}_+(R'G)$ maps R -perfect complexes to R' -perfect complexes (Definition 2.19) and sends $\mathcal{P}(G; R)^\natural$ into $\mathcal{P}(G; R')^\natural$.*

Proof. We have $\text{Res}_1^G(R' \otimes_R^{\mathbb{L}} X) \cong R' \otimes_R^{\mathbb{L}} \text{Res}_1^G(X)$ and $R' \otimes_R^{\mathbb{L}} -: \text{D}(R) \rightarrow \text{D}(R')$ preserves perfect complexes. And if $P \rightarrow X$ is an m -free permutation resolution (over RG), then so is $R' \otimes_R P \cong R' \otimes_R^{\mathbb{L}} P \rightarrow R' \otimes_R^{\mathbb{L}} X$ (over $R'G$). $\square$

2.25. *Remark.* One can also tensor over $\mathbb{Z}$ and use diagonal G -action to get

$$- \otimes_{\mathbb{Z}} -: \text{Mod}(\mathbb{Z}G) \times \text{Mod}(RG) \longrightarrow \text{Mod}(RG)$$

and its derived version $- \otimes_{\mathbb{Z}}^{\mathbb{L}} -: D_+(\mathbb{Z}G) \times D_+(RG) \rightarrow D_+(RG)$. This generalized tensor is composed of the tensors of Propositions 2.23 for R and 2.24 for $\mathbb{Z} \rightarrow R$:

$$X \otimes_{\mathbb{Z}}^{\mathbb{L}} Y \cong (R \otimes_{\mathbb{Z}}^{\mathbb{L}} X) \otimes_R^{\mathbb{L}} Y.$$

It follows in particular that we still have $\mathcal{P}(G; \mathbb{Z})^{\natural} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathcal{P}(G; R)^{\natural} \subseteq \mathcal{P}(G; R)^{\natural}$.

3. ESSENTIAL IMAGE OF THE FUNCTOR $\bar{F}$

Recall the thick subcategory of $D_b(RG)$ from the previous section (Remark 2.13):

$$(3.1) \quad \mathcal{P}(G; R)^{\natural} = \left\{ X \in D_b(RG) \mid \begin{array}{l} X \oplus \Sigma X \text{ admits } m\text{-free permutation} \\ \text{resolutions (Def. 2.1) for all } m \geq 0 \end{array} \right\}.$$

In this section, we prove that the canonical functor $\bar{F}$ of (1.2) in the Introduction is fully-faithful, with $\mathcal{P}(G; R)^{\natural}$ as essential image. See Theorem 3.21. We also prove that $\bar{F}$ is essentially surjective for R a field (Corollary 3.11) or $R = \mathbb{Z}$ (Lemma 3.12). This fact holds for all regular rings: see Corollary 6.20 or Scholium 3.15.

First, we state an abstract form of Carlson [Car00], to reduce several problems to elementary abelian groups, without repeating ourselves. (See Remark 2.25 for $\otimes_{\mathbb{Z}}^{\mathbb{L}}$.)

3.2. Lemma. *Suppose given for every subgroup $H \leq G$ two thick subcategories $\mathcal{C}_1(H)$ and $\mathcal{C}_2(H)$ of $D_b(RH)$ satisfying the following two conditions:*

- (1) *The family $\mathcal{C}_1(-)$ is closed under restriction and $\mathcal{C}_2(-)$ under induction.*
- (2) *For every elementary abelian subgroup $E \leq G$ and $\mathbb{Z}$ -free $\mathbb{Z}E$ -modules W , we have $W \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathcal{C}_1(E) \subseteq \mathcal{C}_2(E)$, meaning $W \otimes_{\mathbb{Z}}^{\mathbb{L}} X \in \mathcal{C}_2(E)$ for all $X \in \mathcal{C}_1(E)$.*

Then we have $\mathcal{C}_1(G) \subseteq \mathcal{C}_2(G)$.

Proof. Let us use the same notation as [Car00, Theorem 2.1]. There exist a $\mathbb{Z}$ -free $\mathbb{Z}G$ -module V , a filtration by $\mathbb{Z}G$ -submodules (for $\mathbb{Z}$ with trivial G -action)

$$0 = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_{\tau} = \mathbb{Z} \oplus V,$$

elementary abelian $E_i \leq G$ and $\mathbb{Z}$ -free $\mathbb{Z}E_i$ -modules W_i , for $1 \leq i \leq \tau$, such that $L_i/L_{i-1} \simeq \text{Ind}_{E_i}^G W_i$. Given now $X \in \text{Ch}_b(RG)$, define in $\text{Ch}_b(RG)$ the objects

$$X' = V \otimes_{\mathbb{Z}} X \quad \text{and} \quad N_i = L_i \otimes_{\mathbb{Z}} X \text{ for all } 0 \leq i \leq \tau.$$

Then, by flatness over $\mathbb{Z}$ of the modules L_i and V , we get a filtration

$$0 = N_0 \subseteq N_1 \subseteq \cdots \subseteq N_{\tau-1} \subseteq N_{\tau} = X \oplus X'$$

by subcomplexes such that (using $\mathbb{Z}$ -flatness of W_i in the third isomorphism)

$$N_i/N_{i-1} \simeq (\text{Ind}_{E_i}^G W_i) \otimes_{\mathbb{Z}} X \cong \text{Ind}_{E_i}^G (W_i \otimes_{\mathbb{Z}} \text{Res}_{E_i}^G(X)) \cong \text{Ind}_{E_i}^G (W_i \otimes_{\mathbb{Z}}^{\mathbb{L}} \text{Res}_{E_i}^G(X)).$$

If $X \in \mathcal{C}_1(G)$, it follows from (1) and (2) that each N_i/N_{i-1} belongs to $\mathcal{C}_2(G)$ and therefore so does X since $\mathcal{C}_2(G)$ is triangulated and thick in $D_b(RG)$. $\square$

3.3. Corollary. *Fix the ring R and consider the following property of G :*

$$(*_G) \quad \mathcal{P}(G; R)^{\natural} = D_b(RG).$$

*If $(*_E)$ holds for all elementary abelian subgroups E of G , then $(*_G)$ holds for G .*

Proof. Use $\mathcal{C}_1(H) = D_b(RH)$, $\mathcal{C}_2(H) = \mathcal{P}(H; R)^{\natural}$ and Proposition 2.18. $\square$

We now study the effect of inverting the order of $|G|$ (cf. Proposition 2.18 (c)).

3.4. Lemma. *Let $r \in R$ and set $R' = R[1/r]$. The canonical functor $R' \otimes_R - : \text{Ch}_b(\text{perm}(G; R)) \rightarrow \text{Ch}_b(\text{perm}(G; R'))$ is essentially surjective.*

Proof. Pick $P' \in \text{Ch}_b(\text{perm}(G; R'))$. We can assume $P_i = 0$ unless $0 \leq i \leq n$. Each $P'_i = R'(A_i)$ for some finite G -set A_i clearly comes from $R(A_i)$ over R . Let $N \geq 1$ and consider the following construction (note the changing powers of r , vertically):

$$\begin{array}{ccccccccccc} P(N) := & \cdots & 0 & \longrightarrow & P'_n & \xrightarrow{\partial_n} & P'_{n-1} & \xrightarrow{\partial_{n-1}} & \cdots & \xrightarrow{\partial_2} & P'_1 & \xrightarrow{\partial_1} & P'_0 & \longrightarrow & 0 \cdots \\ & & \downarrow s(N) & & \downarrow r^{nN} & & \downarrow r^{(n-1)N} & & & & \downarrow r^N & & \downarrow 1 & & \\ P' = & \cdots & 0 & \longrightarrow & P'_n & \xrightarrow{\partial'_n} & P'_{n-1} & \xrightarrow{\partial'_{n-1}} & \cdots & \xrightarrow{\partial'_2} & P'_1 & \xrightarrow{\partial'_1} & P'_0 & \longrightarrow & 0 \cdots \end{array}$$

where $\partial_i := r^N \partial'_i$. This is an isomorphism $s(N): P(N) \xrightarrow{\sim} P'$ in $\text{Ch}_b(\text{perm}(G; R'))$. Increasing $N \gg 1$, one easily arranges that the maps ∂_i in $P(N)$ are defined over R , that they are RG -linear and finally that they form a complex, for all these properties only involve finitely many denominators. Then $P(N)$ provides a source of P' . $\square$

3.5. Proposition. *Let $r \in R$ and set $R' = R[1/r]$. Let $X \in \text{D}_b(RG)$ such that $R' \otimes_R^\mathbb{L} X \in \mathcal{P}(G; R')$. Then there exists an exact triangle $P \rightarrow X \rightarrow T \rightarrow \Sigma P$ in $\text{D}_b(RG)$ where $P \in \text{Ch}_b(\text{perm}(G; R))$ and T is r -torsion, i.e. $r^n \cdot \text{id}_T = 0$ for $n \gg 1$.*

Proof. By Lemma 3.4, there exists $P \in \text{Ch}_b(\text{perm}(G; R))$ and an isomorphism $R' \otimes_R P \xrightarrow{\sim} R' \otimes_R X$ in $\text{D}_b(R'G)$. By the usual localization sequence

$$\text{D}_b^{r\text{-tors}}(RG) \hookrightarrow \text{D}_b(RG) \xrightarrow{R' \otimes_R^\mathbb{L} -} \text{D}_b(R'G)$$

(see Keller [Kel99, Lemma 1.15]), we have $\text{D}_b(R'G) = \text{D}_b(RG)[1/r]$. So the isomorphism $R' \otimes_R P \xrightarrow{\sim} R' \otimes_R X$ is given by a fraction $P \xleftarrow{r^n} P \xrightarrow{f} X$ in $\text{D}_b(RG)$ for some $n \gg 1$ and some $f: P \rightarrow X$ such that $T := \text{cone}(f)$ belongs to $\text{D}_b^{r\text{-tors}}(RG)$. $\square$

3.6. Corollary. *Let G be a p -group and $X \in \text{D}_b(RG)$ be R -perfect (Definition 2.19). Then there exists an exact triangle in $\text{D}_b(RG)$*

$$P \rightarrow X \oplus \Sigma X \rightarrow T \rightarrow \Sigma P$$

where P belongs to $\text{Ch}_b(\text{perm}(G; R))$ and T is p -torsion.

Proof. We apply Proposition 3.5 for $r = p$, so $R' = R[1/p]$. It is easy to check that $X' := R' \otimes_R X \in \text{D}_b(R'G)$ remains R' -perfect. By Corollary 2.20, we have $X \in \mathcal{P}(G; R)^\natural$ hence $X \oplus \Sigma X \in \mathcal{P}(G; R')$. The result then follows by Proposition 3.5. $\square$

3.7. Remark. If X is p -torsion, say $p^n \cdot \text{id}_X = 0$, then the octahedron axiom gives

$$X \in \text{thick}(X \oplus \Sigma X) = \text{thick}(\text{cone}(X \xrightarrow{p^n} X)) \subseteq \text{thick}(\text{cone}(X \xrightarrow{p} X)).$$

Adapting [BB20] let us now see when the trivial RG -module R belongs to $\mathcal{P}(G; R)$.

3.8. Proposition. *The following are equivalent:*

- (i) *The trivial RG -module R belongs to $\mathcal{P}(G; R)$.*
- (ii) *It admits a 0-free permutation resolution (Lemma 2.4): There is a resolution $0 \rightarrow P_n \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow R \rightarrow 0$ by permutation RG -modules with P_0 free.*

Proof. (i) $\Rightarrow$ (ii) is trivial. (ii) $\Rightarrow$ (i) by Proposition 2.22 and $R^{\otimes m} \cong R$, $\forall m \geq 0$. $\square$

3.9. Proposition. *Let G be an abelian group. Then R belongs to $\mathcal{P}(G; R)$.*

Proof. Consider first the cyclic group $C_n = \langle x \mid x^n = 1 \rangle$ of order n , so that $RG = R[x]/(x^n - 1)$. Then the sequence of RC_n -modules

$$0 \longrightarrow R \xrightarrow{(x^{n-1} + \dots + x + 1) \cdot} R[x]/(x^n - 1) \xrightarrow{(x-1) \cdot} R[x]/(x^n - 1) \xrightarrow{x \mapsto 1} R \longrightarrow 0$$

is easily verified to be exact, with both modules R acted upon trivially.

In general, since G is a product of cyclic groups, we find subgroups $H_1, \dots, H_r$ of G such that each G/H_i is cyclic and $H_1 \cap \dots \cap H_r = 1$. Inflating from the cyclic groups G/H_i the above resolution, we get quasi-isomorphisms

$$\begin{array}{ccccccc} Q(i) := & 0 & \longrightarrow & R & \longrightarrow & R(G/H_i) & \longrightarrow & R(G/H_i) & \longrightarrow & 0 \\ s_i \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\ \mathbb{1} = & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & R & \longrightarrow & 0 \end{array}$$

for $i = 1, \dots, r$. Then $s_1 \otimes \dots \otimes s_r: Q(1) \otimes_R \dots \otimes_R Q(r) \rightarrow \mathbb{1}$ gives a 0-free permutation resolution by the Mackey formula. Then apply Proposition 3.8. $\square$

3.10. Lemma. *Let G be an abelian group. Then every bounded complex of permutation RG -modules belongs to $\mathcal{P}(G; R)$.*

Proof. Let $H \leq G$ be a subgroup. By Propositions 3.9 and 2.18, we have $R(G/H) = \text{Ind}_H^G(R) \in \text{Ind}_H^G(\mathcal{P}(H; R)) \subseteq \mathcal{P}(G; R)$. Finally, $\mathcal{P}(G; R)$ is triangulated. $\square$

3.11. Corollary. *Let k be a field. Then $\mathcal{P}(G; k)^\natural = \mathcal{Q}(G; k)^\natural = \text{D}_b(kG)$.*

Proof. By Corollary 3.3, we can assume that G is an elementary abelian p -group. If p is invertible in k , we are done by Corollary 2.20. If $\text{char}(k) = p$ then $k \in \mathcal{P}(G; k)$ since G is abelian (Proposition 3.9) and $\text{thick}(k) = \text{D}_b(kG)$ since G is a p -group. $\square$

3.12. Lemma. *We have $\mathcal{P}(G; \mathbb{Z})^\natural = \text{D}_b(\mathbb{Z}G)$.*

Proof. As before, we reduce to the case of G an elementary abelian p -group by Corollary 3.3, for some prime p . Pick $X \in \text{D}_b(\mathbb{Z}G)$. Note that X is $\mathbb{Z}$ -perfect and, by Corollary 3.6, there exists an exact triangle

$$P \rightarrow X \oplus \Sigma X \rightarrow T \rightarrow \Sigma P$$

in $\text{D}_b(\mathbb{Z}G)$ where P is a complex of permutation modules, and T is p -torsion. In particular $P \in \mathcal{P}(G; \mathbb{Z})$ already, by Lemma 3.10. So it suffices to prove $T \in \mathcal{P}(G; \mathbb{Z})^\natural$, i.e. we can assume that $p^n \cdot X = 0$ for some $n \gg 1$. By Remark 3.7 we then have $X \in \text{thick}(\text{cone}(X \xrightarrow{p} X))$. We compute $\text{cone}(X \xrightarrow{p} X) \cong \text{cone}(\mathbb{Z} \xrightarrow{p} \mathbb{Z}) \otimes_{\mathbb{Z}}^L X \cong i_* \text{Li}^*(X)$ by the projection formula for the adjunction

$$\begin{array}{ccc} & \text{D}_b(\mathbb{Z}G) & \\ \text{Li}^* = \mathbb{F}_p \otimes_{\mathbb{Z}}^L - \downarrow & \uparrow i_* & \\ & \text{D}_b(\mathbb{F}_p G) & \end{array}$$

given by the usual extension and restriction of scalars along $\mathbb{Z} \rightarrow \mathbb{F}_p = \mathbb{Z}/p$. Hence

$$(3.13) \quad X \in i_*(\text{D}_b(\mathbb{F}_p G)).$$

Using that $\text{thick}(\mathbb{F}_p) = \text{D}_b(\mathbb{F}_p G)$ since G is a p -group we continue:

$$(3.14) \quad X \in i_*(\text{thick}(\mathbb{F}_p)) \subseteq \text{thick}(i_*(\mathbb{F}_p)) = \text{thick}(\text{cone}(\mathbb{Z} \xrightarrow{p} \mathbb{Z})) \subseteq \text{thick}(\mathbb{Z}).$$

And we conclude from $\mathbb{Z} \in \mathcal{P}(G; \mathbb{Z})$, as already established in Proposition 3.9. $\square$

3.15. *Scholium.* The above proof *almost* works for any regular ring R instead of $\mathbb{Z}$. Of course $\mathbb{F}_p$ needs to be replaced by $\bar{R} = R/p$. Using regularity, we can go up to (3.13) in the above proof. One point which is more complicated is that $\bar{R}$ does not necessarily generate $D_b(\bar{R}G)$, as we used to pass to (3.14) for $\mathbb{F}_p$. (Here $\bar{R}$ is not necessarily regular.) Instead, for G a p -group, one can show that the image of $D_b(\bar{R})$ under Infl_1^G (i.e. $\bar{R}$ -modules with trivial G -action) generates $D_b(\bar{R}G)$. This uses nilpotence of the augmentation ideal $I := \text{Ker}(\bar{R}G \rightarrow \bar{R})$, and the associated finite filtration $\cdots I^{\ell+1}N \subseteq I^\ell N \cdots$ of any $\bar{R}G$ -module N , in which every $I^\ell N / I^{\ell+1}N$ has trivial G -action. Then one uses regularity of R one more time and the commutativity of the following square

$$(3.16) \quad \begin{array}{ccc} D_b(\bar{R}) & \xrightarrow{i_*} & D_b(R) = D_{\text{perf}}(R) = \text{thick}(R) \\ \text{Infl}_1^G \downarrow & & \downarrow \text{Infl}_1^G \\ D_b(\bar{R}G) & \xrightarrow{i_*} & D_b(RG) \end{array}$$

to conclude via Proposition 3.9 again but with the following replacement of (3.14):

$$X \in \text{thick}(i_*(\text{Infl}_1^G D_b(\bar{R}))) \underset{(3.16)}{\subseteq} \text{thick}(\text{Infl}_1^G R) \underset{\text{Prop. 3.9}}{\subseteq} \mathcal{P}(G; R)^\natural.$$

Returning to arbitrary noetherian commutative rings R , we extend Lemma 3.10.

3.17. **Corollary.** *Each bounded complex of permutation RG -modules is in $\mathcal{P}(G; R)^\natural$.*

Proof. Let $H \leq G$ be a subgroup. By Lemma 3.12, we know that $\mathbb{Z} \in \mathcal{P}(H; \mathbb{Z})^\natural$. By Proposition 2.24 for the unique homomorphism $\mathbb{Z} \rightarrow R$, we get that $R \in \mathcal{P}(H; R)^\natural$. By Proposition 2.18, we deduce that $R(G/H) = \text{Ind}_H^G(R) \in \mathcal{P}(G; R)^\natural$. We then conclude since $\mathcal{P}(G; R)^\natural \subseteq D_b(RG)$ is thick and triangulated (Proposition 2.9). $\square$

We also use later the following technical consequence (see Remark 2.25 for $\otimes_{\mathbb{Z}}^\perp$):

3.18. **Lemma.** *Let $W \in D_b(\mathbb{Z}G)$ and $X \in \mathcal{P}(G; R)^\natural$. Then $W \otimes_{\mathbb{Z}}^\perp X \in \mathcal{P}(G; R)^\natural$.*

Proof. We have $D_b(\mathbb{Z}) \otimes_{\mathbb{Z}}^\perp \mathcal{P}(G; R)^\natural \underset{3.12}{=} \mathcal{P}(G; \mathbb{Z})^\natural \otimes_{\mathbb{Z}}^\perp \mathcal{P}(G; R)^\natural \underset{2.25}{\subseteq} \mathcal{P}(G; R)^\natural$. $\square$

3.19. **Corollary.** *Let X be an object of $D_b(RG)$. The following are equivalent:*

- (i) *The complex X belongs to $\mathcal{P}(G; R)^\natural$.*
- (ii) *For every Sylow subgroup $H \leq G$, the restriction $\text{Res}_H^G(X)$ belongs to $\mathcal{P}(H; R)^\natural$.*
- (iii) *For every elementary abelian subgroup $E \leq G$, we have $\text{Res}_E^G(X) \in \mathcal{P}(E; R)^\natural$.*

Proof. (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are immediate from stability under restriction as in Proposition 2.18 (a). To prove (iii) $\Rightarrow$ (i), we apply Lemma 3.2 to $\mathcal{C}_2(H) = \mathcal{P}(H; R)^\natural$ and

$$\mathcal{C}_1(H) = \{ X \in D_b(RH) \mid \text{Res}_E^H(X) \in \mathcal{P}(E; R)^\natural \text{ for all elementary abelian } E \leq H \}$$

for $H \leq G$. Condition (1) is satisfied by Proposition 2.18. To verify Condition (2), let $E \leq G$ be elementary abelian. In that case $\mathcal{C}_1(E) = \mathcal{C}_2(E)$ and Condition (2) holds by Lemma 3.18. Hence Lemma 3.2 tells us $\mathcal{C}_1(G) \subseteq \mathcal{C}_2(G)$. $\square$

3.20. *Remark.* In Corollary 3.19, once we know that $\text{Res}_1^G(X)$ is perfect, it suffices to check (ii) or (iii) for the p -subgroups H or E for those primes p such that $X \xrightarrow{p} X$ is not invertible (in particular not invertible in R). This holds by Corollary 2.20.

We now give a more conceptual description of the subcategory $\mathcal{P}(G; R)^\natural$ of (3.1):

3.21. Theorem. *Let G be a finite group and R a commutative noetherian ring. The canonical functor $\bar{F}$ of (1.2) restricts to an exact equivalence*

$$(3.22) \quad \bar{F}: \left(\frac{K_b(\text{perm}(G; R))}{K_{b,ac}(\text{perm}(G; R))} \right)^{\natural} \xrightarrow{\simeq} \mathcal{P}(G; R)^{\natural}.$$

In particular, $\mathcal{P}(G; R)^{\natural}$ is the essential image of $\bar{F}$ in $D_b(RG)$.

Proof. By Corollary 3.17, the canonical functor $K_b(\text{perm}(G; R)) \rightarrow D_b(\text{mod}(RG))$ lands inside $\mathcal{P}(G; R)^{\natural}$. Since the latter is idempotent-complete there exists a well-defined exact functor $\bar{F}$ as in (3.22). By Definition 2.1 (for $m = 0$), it is clear that $\bar{F}$ is surjective-up-to-direct-summands, hence it suffices to prove that the functor

$$\bar{\mathcal{K}} := \frac{K_b(\text{perm}(G; R))}{K_{b,ac}(\text{perm}(G; R))} \xrightarrow{\bar{F}} D_b(RG)$$

is fully faithful. As every X is a retract of $X \oplus \Sigma X$, it suffices to prove that the homomorphism $\bar{F}: \text{Hom}_{\bar{\mathcal{K}}}(X \oplus \Sigma X, Y) \rightarrow \text{Hom}_{D_b(RG)}(X \oplus \Sigma X, Y)$ is an isomorphism for every $X, Y \in K_b(\text{perm}(G; R))$. By Corollary 3.17, we know that such complex X belongs to $\mathcal{P}(G; R)^{\natural}$ hence $X \oplus \Sigma X \in \mathcal{P}(G; R)$. So it suffices to show that for every $X, Y \in K_b(\text{perm}(G; R))$ such that $X \in \mathcal{P}(G; R)$, the homomorphism

$$\bar{F}: \text{Hom}_{\bar{\mathcal{K}}}(X, Y) \rightarrow \text{Hom}_{D_b(RG)}(X, Y)$$

is an isomorphism. For surjectivity, let $f s^{-1}: X \rightarrow Y$ be represented by a fraction $X \xleftarrow{s} Z \xrightarrow{f} Y$ in $\text{Ch}_b(RG)$ where s is a quasi-isomorphism. Since X belongs to $\mathcal{P}(G; R)$, so does Z by Proposition 2.7. So Z admits a 0-free permutation resolution, i.e. there exists a quasi-isomorphism $t: P \rightarrow Z$ with $P \in \text{Ch}_b(\text{perm}(G; R))$. Hence our morphism $f s^{-1} = (ft)(st)^{-1}$ comes from $X \xleftarrow{st} P \xrightarrow{ft} Y$ in $\text{Hom}_{\bar{\mathcal{K}}}(X, Y)$. Injectivity is similar (or follows from conservativity and fullness of $\bar{F}$). $\square$

4. BIG SINGULARITY CATEGORY

The target of our ‘cohomological singularity’ functor $\chi^G: D(RG) \rightarrow D^{\text{sing}}(R)$ is the big singularity category of the coefficient ring R . Let us remind the reader.

4.1. Recollection. As in [Kra05], let $\mathcal{A}$ be a locally noetherian Grothendieck category compactly generated derived category. For $\mathcal{A} = \text{Mod}(R)$, the noetherian objects $\text{noeth } \mathcal{A}$ is $\text{mod}(R)$ and $D(\mathcal{A})$ is generated by $D(\mathcal{A})^c = D_{\text{per}}(R)$. Similarly for $\mathcal{A} = \text{Mod}(RG)$. The *big singularity category* (or *stable derived category*) of $\mathcal{A}$ is

$$D^{\text{sing}}(\mathcal{A}) = K_{ac}(\text{Inj } \mathcal{A}),$$

the full subcategory of the big homotopy category of injectives $K(\text{Inj } \mathcal{A})$ spanned by acyclic complexes. There is a recollement $Q_{\lambda} \dashv Q \dashv Q_{\rho}$ and $I_{\lambda} \dashv I \dashv I_{\rho}$

$$(4.2) \quad \begin{array}{c} D(\mathcal{A}) \\ \begin{array}{ccc} Q_{\lambda} \downarrow & \uparrow \downarrow & \downarrow Q_{\rho} \\ & Q & \\ & \downarrow & \end{array} \\ K(\text{Inj } \mathcal{A}) \\ \begin{array}{ccc} I_{\lambda} \downarrow & \uparrow \downarrow & \downarrow I_{\rho} \\ & I & \\ & \downarrow & \end{array} \\ D^{\text{sing}}(\mathcal{A}) = K_{ac}(\text{Inj } \mathcal{A}) \end{array}$$

for $I: K_{ac}(\text{Inj } \mathcal{A}) \hookrightarrow K(\text{Inj } \mathcal{A})$ the inclusion and $Q: K(\text{Inj } \mathcal{A}) \twoheadrightarrow D(\mathcal{A})$ the localization.

The *singularity functor* (Krause's *stabilization functor*) is defined as

$$\text{sing}_{\mathcal{A}} : \mathcal{D}(\mathcal{A}) \xrightarrow{I_{\lambda} \circ Q_{\rho}} \mathcal{D}^{\text{sing}}(\mathcal{A}).$$

There is a natural transformation $Q_{\lambda} \rightarrow Q_{\rho}$ that is invertible on compacts:

$$(4.3) \quad Q_{\lambda}(X) \cong Q_{\rho}(X) \quad \text{if } X \in \mathcal{D}(\mathcal{A})^c$$

by [Kra05, Lemma 5.2]. On the larger subcategory $\mathcal{D}_b(\text{noeth } \mathcal{A})$, the right adjoint Q_{ρ} defines an inverse to the equivalence of [Kra05, § 2] identifying $\mathcal{K}(\text{Inj } \mathcal{A})^c$

$$Q : \mathcal{K}(\text{Inj } \mathcal{A})^c \xrightarrow{\sim} \mathcal{D}_b(\text{noeth } \mathcal{A}).$$

In summary, we have a finite localization sequence $\mathcal{D}(\mathcal{A}) \xrightarrow{Q_{\lambda}} \mathcal{K}(\text{Inj } \mathcal{A}) \xrightarrow{I_{\lambda}} \mathcal{D}^{\text{sing}}(\mathcal{A})$ as in (4.2) and the triangulated category $\mathcal{D}^{\text{sing}}(\mathcal{A})$ is compactly generated with compact part the (usual) *small singularity category*, idempotent-completed

$$\mathcal{D}^{\text{sing}}(\mathcal{A})^c \cong \left(\frac{\mathcal{K}(\text{Inj } \mathcal{A})^c}{(Q_{\lambda} \mathcal{D}(\mathcal{A}))^c} \right)^{\natural} \cong \left(\frac{\mathcal{D}_b(\text{noeth } \mathcal{A})}{\mathcal{D}(\mathcal{A})^c} \right)^{\natural} = \mathcal{D}_b^{\text{sing}}(\mathcal{A})^{\natural}.$$

Finally, we note that since $\mathcal{K}(\text{Inj } \mathcal{A})$ is compactly generated and since the inclusion $\mathcal{K}(\text{Inj } \mathcal{A}) \hookrightarrow \mathcal{K}(\mathcal{A})$, that we denote J , preserves products and coproducts (because $\mathcal{A}$ is locally noetherian), there is another useful triple of adjoints $J_{\lambda} \dashv J \dashv J_{\rho}$:

$$(4.4) \quad \begin{array}{ccc} & \mathcal{K}(\mathcal{A}) & \\ J_{\lambda} \downarrow & \uparrow J & \downarrow J_{\rho} \\ & \mathcal{K}(\text{Inj } \mathcal{A}) & \end{array}$$

4.5. Lemma. *Keep the above notation, e.g. for $\mathcal{A} = \text{Mod}(RG)$. Let X be in $\mathcal{K}(\mathcal{A})$.*

- (a) *The object $Q_{\rho}(X) \in \mathcal{K}(\text{Inj } \mathcal{A})$ in (4.2) is a K -injective resolution of X . In particular, if $X \in \mathcal{K}_{-}(\mathcal{A})$ is left-bounded then $Q_{\rho}(X)$ belongs to $\mathcal{K}_{-}(\text{Inj } \mathcal{A})$.*
- (b) *If $X \in \mathcal{K}_{-}(\mathcal{A})$ is left-bounded, the object $J_{\lambda}(X) \in \mathcal{K}(\text{Inj } \mathcal{A})$ in (4.4) is an injective resolution of X . Hence if $X \in \mathcal{K}_{-, \text{ac}}(\mathcal{A})$ is also acyclic then $J_{\lambda}(X) = 0$.*
- (c) *The restrictions of the two functors Q_{ρ} and J_{λ} to $\mathcal{K}_{-}(\mathcal{A})$ are isomorphic.*

Proof. By [Kra05, Remark 3.7], we have $Q J_{\lambda} \cong Q^{+}$, where $Q^{+} : \mathcal{K}(\mathcal{A}) \rightarrow \mathcal{D}(\mathcal{A})$ is the (Bousfield) localization defining $\mathcal{D}(\mathcal{A})$. It follows that $J Q_{\rho}$ is right adjoint to Q^{+} . So, if we let $i := J Q_{\rho} \circ Q^{+}$, every $X \in \mathcal{K}(\mathcal{A})$ fits in an exact triangle

$$(4.6) \quad a(X) \rightarrow X \xrightarrow{\eta} i(X) \rightarrow \Sigma a(X)$$

in $\mathcal{K}(\mathcal{A})$, where $a(X) \in \text{Ker}(Q^{+}) = \mathcal{K}_{\text{ac}}(\mathcal{A})$ and $i(X)$ belongs to $\text{Ker}(Q^{+})^{\perp} = \mathcal{K}_{\text{ac}}(\mathcal{A})^{\perp} =: \mathcal{K}_{\text{Inj}}(\mathcal{A})$, that is, $i(X)$ is K -injective by definition. In other words, (4.6) is the essentially unique triangle providing the K -injective resolution of X (see [Kra05, Corollary 3.9] if necessary). Suppressing the functors J and Q^{+} that are just the identity on objects, we have $i(X) = Q_{\rho}(X)$, which gives (a).

Let now $A \in \mathcal{K}_{-, \text{ac}}(\mathcal{A})$. The unit $\eta' : A \rightarrow J J_{\lambda}(A)$ is a map from a left-bounded acyclic to a complex of injectives, hence $\eta' = 0$ in $\mathcal{K}(\mathcal{A})$. But $J_{\lambda}(\eta') : J_{\lambda} \xrightarrow{\sim} J_{\lambda} J J_{\lambda}$ is invertible (J being fully faithful). Thus $J_{\lambda}(A) = 0$, as in the second claim of (b).

Take now $X \in \mathcal{K}_{-}(\mathcal{A})$ arbitrary and an injective resolution $i(X) \in \mathcal{K}_{-}(\text{Inj } \mathcal{A})$. There is a triangle (4.6) with $a(X)$ exact and left-bounded since X and $i(X)$ are. By the above for $A = a(X)$, we already know that $J_{\lambda}(a(X)) = 0$. Applying J_{λ} to the triangle (4.6) we get $J_{\lambda}(X) \cong J_{\lambda}(i(X)) \cong i(X)$ since $i(X) \in \mathcal{K}(\text{Inj } \mathcal{A})$. Hence (b).

Part (c) is now immediate from the uniqueness of K -injective resolutions. $\square$

Let us now specialize to $\mathcal{A} = \text{Mod}(R)$.

4.7. Lemma. *Let X be an object of $\text{D}(R)$. Then $\text{sing}(X) = 0$ in $\text{D}^{\text{sing}}(R)$ if and only if $Q_\rho(X)$ belongs to the localizing subcategory of $\text{K}(\text{Inj}(R))$ generated by $Q_\rho(R)$.*

Proof. We have $\text{sing} = I_\lambda \circ Q_\rho$ by definition. So we have $\text{sing}(X) = 0$ if and only if $Q_\rho(X) \in \text{Ker}(I_\lambda)$ and by (4.2) that kernel is $\text{Ker}(I_\lambda) = Q_\lambda(\text{D}(R)) = Q_\lambda(\text{Loc}(R)) = \text{Loc}(Q_\lambda(R))$, where the last equality holds since Q_λ is coproduct-preserving and fully-faithful. We conclude from $Q_\lambda(R) \cong Q_\rho(R)$ since $R \in \text{D}(R)^c$; see (4.3). $\square$

4.8. Remark. In fact, $\text{sing}(X) = 0$ is also equivalent to $Q_\lambda(X) \xrightarrow{\sim} Q_\rho(X)$.

We now describe a class of not necessarily perfect complexes in $\text{Ker}(\text{sing})$. As Q_ρ and $\text{sing} = I_\lambda \circ Q_\rho$ may not commute with coproducts, we need to be careful.

4.9. Remark. We say that a family of complexes $\{X(i)\}_{i \in I}$ in $\text{Ch}(R)$ is *degreewise finite* if for every $n \in \mathbb{Z}$, the set $\{i \in I \mid X(i)_n \neq 0\}$ is finite. Since (co)products in $\text{D}(R)$ are computed degreewise, the canonical map $\coprod_{i \in I} X(i) \rightarrow \prod_{i \in I} X(i)$ is an isomorphism in that case and *only then* do we write $\oplus_{i \in I} X(i)$ to refer to both coproduct and product. If all $X(i)$ are complexes of injectives, then so is $\oplus_{i \in I} X(i)$.

4.10. Definition. An object $X \in \text{D}(R)$ is *weakly perfect* if there exists an isomorphism $X \simeq \oplus_{i \in \mathbb{N}} P(i)$ in $\text{D}(R)$ for a countable family $\{P(i)\}_{i \in \mathbb{N}}$ of strictly perfect complexes $P(i) \in \text{Ch}_b(\text{proj}(R))$ such that $P(i)_n = 0$ unless $a_i \geq n \geq b_i$ and such that $\lim_{i \rightarrow \infty} a_i = -\infty$. (Such a family $\{P(i)\}_{i \in \mathbb{N}}$ is necessarily degreewise finite.)

4.11. Example. For any P perfect, the complex $X = \coprod_{i \geq 0} \Sigma^{-i} P$ is weakly perfect.

4.12. Lemma. *Let $X \in \text{D}(R)$ be weakly perfect. Then $\text{sing}(X) = 0$ in $\text{D}^{\text{sing}}(R)$.*

Proof. Let $X \simeq \oplus_{i \in \mathbb{N}} P(i)$ for $\{P(i)\}_{i \in \mathbb{N}}$ as in Definition 4.10. By Lemma 4.5 (a), $P(i) \rightarrow Q_\rho(P(i))$ are injective resolutions and thus live in degrees $\leq a_i$. So the family $\{Q_\rho(P(i))\}_{i \in \mathbb{N}}$ is degreewise finite (Remark 4.9) and $\coprod_{i \in \mathbb{N}} Q_\rho(P(i)) \cong \prod_{i \in \mathbb{N}} Q_\rho(P(i))$ in $\text{K}(\text{Inj}(R))$. Since the right adjoint Q_ρ preserves products, we get

$$Q_\rho(X) \simeq Q_\rho\left(\coprod_{i \in \mathbb{N}} P(i)\right) \cong \prod_{i \in \mathbb{N}} Q_\rho(P(i)) \cong \prod_{i \in \mathbb{N}} Q_\rho(P(i)).$$

Hence $Q_\rho(X)$ belongs to $\text{Loc}(Q_\rho(\text{D}_{\text{perf}}(R)))$ and $\text{sing}(X) = 0$ by Lemma 4.7. $\square$

4.13. Lemma. *Let $X, Y \in \text{D}(R)$ be weakly perfect. Then $X \otimes_R^L Y$ is weakly perfect.*

Proof. Say $X \simeq \oplus_{i \in \mathbb{N}} P(i)$ and $Y \simeq \oplus_{j \in \mathbb{N}} Q(j)$ with $P(i)$ and $Q(j)$ strictly perfect complexes, respectively concentrated in degrees between a_i and b_i and between c_j and d_j , and such that $\lim_i a_i = \lim_j c_j = -\infty$. Then $X \otimes_R^L Y \simeq \coprod_{i,j} P(i) \otimes_R Q(j)$, each $P(i) \otimes_R Q(j)$ is strictly perfect and concentrated in degrees between $a_i + c_j$ and $b_i + d_j$. As $\mathbb{N} \times \mathbb{N}$ is countable and for each $n \in \mathbb{Z}$ the set $\{(i, j) \in \mathbb{N} \times \mathbb{N} \mid a_i + c_j \geq n\}$ is finite (otherwise $\lim_i a_i = \lim_j c_j = -\infty$ would fail), we get the result. $\square$

5. COHOMOLOGICAL SINGULARITY

In this section, we define the announced cohomological singularity functor (1.5).

5.1. Recollection. The functor that equips every R -module with trivial G -action

$$\text{Infl}_1^G \cong \text{hom}_R(R, -) \cong R \otimes_R - : \text{Mod}(R) \rightarrow \text{Mod}(RG).$$

has adjoints the usual G -orbits $(-)_G$ and G -fixed points $(-)^G$

$$(5.2) \quad R \otimes_{RG} - = (-)_G \quad \left(\begin{array}{c} \text{Mod}(RG) \\ \uparrow \text{Infl}_1^G \\ \text{Mod}(R) \end{array} \right) \quad \text{hom}_{RG}(R, -) = (-)^G$$

This triple of adjoints passes to homotopy categories of complexes on the nose. For derived categories, we left-derive the left adjoint and right-derive the right one:

$$(5.3) \quad R \otimes_{RG}^L - =: (-)_{hG} \quad \left(\begin{array}{c} D(RG) \\ \uparrow \text{Infl}_1^G \\ D(R) \end{array} \right) \quad \text{Rhom}_{RG}(R, -) =: (-)^{hG}$$

So $(-)^{hG}$ provides a complex whose homology groups are G -cohomology as in (1.6).

5.4. Definition. Let $H \leq G$ be a subgroup. The H -cohomological singularity functor $\chi^H = \text{sing}_R \circ (-)^{hH}$ is the following composite (see Recollection 4.1 for sing):

$$\chi^H : D(RG) \xrightarrow{\text{Res}_H^G} D(RH) \xrightarrow{(-)^{hH}} D(R) \xrightarrow{\text{sing}_R} D^{\text{sing}}(R).$$

We say that a complex $X \in D(RG)$ is H -cohomologically perfect if $\chi^H(X) = 0$. We say that X is cohomologically perfect, if it is H -cohomologically perfect for all subgroups $H \leq G$, that is, if $\oplus_H \chi^H(X) = 0$ in $D^{\text{sing}}(R)$.

5.5. Remark. We remind the reader that although $\text{Ker}(\text{sing}) \cap D_b(RG) = D_{\text{perf}}(R)$, the kernel of $\text{sing} : D(R) \rightarrow D^{\text{sing}}(R)$ on the big derived category is larger than $D_{\text{perf}}(R)$; see for instance Lemma 4.12. So even when $H = 1$, being 1-cohomologically perfect is more flexible than being R -perfect (Definition 2.19) although the two notions coincide when $X \in D_b(RG)$ is bounded, i.e. when $\text{Res}_1^G(X) \in D_b(R)$.

For more general subgroups $H \leq G$, even a bounded complex $X \in D_b(RG)$ can be H -cohomologically perfect without X^{hH} being perfect; see Example 5.9.

We provide a better justification of the terminology in Remark 6.19.

5.6. Remark. The functor $(-)^G \cong \text{hom}_{RG}(R, -)$ is a special value of the bifunctor

$$\text{Mod}(RG)^{\text{op}} \times \text{Mod}(RG) \xrightarrow{\text{hom}_{RG}(-, -)} \text{Mod}(R).$$

It follows that for any $X \in D(RG)$ the object X^{hG} is represented by both

$$(5.7) \quad \text{hom}_{RG}(P_R, X) \quad \text{and} \quad \text{hom}_{RG}(R, i(X))$$

where $P_R \rightarrow R$ is a projective resolution of R as an RG -module, and $X \rightarrow i(X)$ is a K-injective resolution of X , for both are quasi-isomorphic to $\text{hom}_{RG}(P_R, i(X))$.

5.8. Remark. If the order $|G|$ is invertible in R , then the trivial RG -module R is projective by (2.17). In that case, $(-)^G$ is exact and coincides with $(-)^{hG}$.

We can use this to see that being G -cohomologically perfect does not imply being H -cohomologically perfect for each subgroup $H \leq G$, even for $H = 1$.

5.9. Example. Let $R = \mathbb{Z}/9$ and $G = C_2 = \langle x \mid x^2 = 1 \rangle$. Consider the R -module $M = \mathbb{Z}/3$ with the action of x by -1 . We have $M^G = 0$ hence $M^{hG} = 0$ by Remark 5.8. In particular, M is G -cohomologically perfect but it is not H -cohomologically perfect for the subgroup $H = 1$ since $\mathbb{Z}/3$ is not perfect over $\mathbb{Z}/9$.

Let us establish some generalities about the cohomological singularity functor.

5.10. Proposition. *Let $H \leq G$ be a subgroup. There are canonical isomorphisms*

$$(-)^{\mathrm{h}G} \circ \mathrm{Ind}_H^G \cong (-)^{\mathrm{h}H} \quad \text{and} \quad \chi^G \circ \mathrm{Ind}_H^G \cong \chi^H.$$

Proof. This first follows from the relation $\mathrm{Res}_H^G \circ \mathrm{Infl}_1^G = \mathrm{Infl}_1^H$, by taking right adjoints and right-deriving. The second follows by post-composing with sing_R . $\square$

5.11. Corollary. *Let $H \leq G$. Then induction $\mathrm{Ind}_H^G: \mathrm{D}_b(RH) \rightarrow \mathrm{D}_b(RG)$ and restriction $\mathrm{Res}_H^G: \mathrm{D}_b(RG) \rightarrow \mathrm{D}_b(RH)$ preserve cohomologically perfect complexes.*

Proof. Restriction is built into Definition 5.4. For induction, it follows immediately from the Mackey formula and Proposition 5.10. $\square$

5.12. Lemma. *Let G be an elementary abelian group and $X \in \mathcal{P}(G; R)^\natural$; see (3.1). Then X is cohomologically perfect.*

Proof. By Theorem 3.21, it suffices to prove that $R(G/H)$ is cohomologically perfect for every $H \leq G$. By Mackey and Proposition 5.10, it suffices to show $\chi^G(R) = 0$.

Assume first that $G = C_p = \langle x \mid x^p = 1 \rangle$ is cyclic of order p . We compute $R^{\mathrm{h}G}$ using (5.7). Let P_R be the usual projective resolution of R :

$$(5.13) \quad \cdots \rightarrow RC_p \xrightarrow{x-1} RC_p \xrightarrow{x^{p-1}+\cdots+1} RC_p \xrightarrow{x-1} RC_p \rightarrow 0$$

Applying $\mathrm{hom}_{RC_p}(-, R)$, we conclude that $R^{\mathrm{h}G}$ is represented by the complex

$$0 \rightarrow R \xrightarrow{0} R \xrightarrow{p} R \xrightarrow{0} R \xrightarrow{p} R \xrightarrow{0} R \rightarrow \cdots,$$

which is weakly perfect in the sense of Definition 4.10.

In general, let $G = C_p^{\times r}$. Then $RG = (RC_p) \otimes_R \cdots \otimes_R (RC_p)$, with r factors.

Let $P_R^{(i)}$ be the inflation of (5.13) along the projection $G \rightarrow C_p$ onto the i th factor. One checks that one has a canonical isomorphism of complexes of R -modules

$$\mathrm{hom}_{RG}(P_R^{(1)}, R) \otimes_R \cdots \otimes_R \mathrm{hom}_{RG}(P_R^{(r)}, R) \xrightarrow{\sim} \mathrm{hom}_{RG}(P_R^{(1)} \otimes_R \cdots \otimes_R P_R^{(r)}, R).$$

The tensor product $\otimes_{i=1}^r P_R^{(i)}$ on the right-hand side is a free resolution of R as an RG -module hence the right-hand side computes $R^{\mathrm{h}G}$. As $\mathrm{hom}_{RG}(P_R^{(i)}, R) \cong \mathrm{hom}_{RC_p}(P_R, R)$ is weakly perfect over R by the above special case, we conclude via Lemma 4.13 that $R^{\mathrm{h}G}$ is weakly perfect. Thus $\mathrm{sing}(R^{\mathrm{h}G}) = 0$ by Lemma 4.12. $\square$

We can now extend Lemma 5.12 to any finite group G .

5.14. Proposition. *Every object of $\mathcal{P}(G; R)^\natural$ is cohomologically perfect.*

Proof. We apply Lemma 3.2 with $\mathcal{C}_1(G) = \mathcal{P}(G; R)^\natural$ and $\mathcal{C}_2(G)$ the thick subcategory of $\mathrm{D}_b(RG)$ spanned by cohomologically perfect complexes. The former is closed under restriction by Proposition 2.18, and the latter is closed under induction by Corollary 5.11. This verifies Condition (1) of Lemma 3.2. For Condition (2), let $E \leq G$ be elementary abelian and W a $\mathbb{Z}$ -free $\mathbb{Z}E$ -module and $X \in \mathcal{C}_1(E) = \mathcal{P}(E; R)^\natural$. By Lemma 3.18, we have $W \otimes_{\mathbb{Z}} X \in \mathcal{P}(E; R)^\natural \subseteq \mathcal{C}_2(E)$ where the latter inclusion holds by Lemma 5.12. Hence Condition (2) holds too. $\square$

5.15. Corollary. *Every bounded complex X of permutation RG -modules is cohomologically perfect.*

Proof. By Corollary 3.17 we have $X \in \mathcal{P}(G; R)^\natural$ and then use Proposition 5.14. $\square$

We can now apply Proposition 5.14 to show that being R -perfect (Definition 2.19) is not sufficient to belong to $\text{Im}(\bar{F}) = \mathcal{P}(G; R)^\natural$. Compare Corollary 2.20.

5.16. *Example.* Let $k = \mathbb{F}_2$ and consider the ring $R = k[x]/\langle x^2 - 1 \rangle$. Take $G = C_2 = \langle y \mid y^2 = 1 \rangle$ cyclic of order 2. Let $X = R_x$ denote the ring R viewed as an RG -module with the non-trivial action of y via x . This $X \in \text{D}_b(RC_2)$ is R -perfect but we claim that $\chi^{C_2}(X) \neq 0$. As RC_2 is self-injective, the following resolution

$$0 \rightarrow R_x \xrightarrow{y-x} RC_2 \xrightarrow{y-x} RC_2 \rightarrow \cdots$$

is an injective resolution of R_x . Computing $(R_x)^{\text{h}G}$ in $\text{D}(R)$ as in (5.7) with the above $i(X) = \cdots 0 \rightarrow RC_2 \xrightarrow{y-x} RC_2 \xrightarrow{y-x} RC_2 \rightarrow \cdots$, we get that $(R_x)^{\text{h}G}$ is

$$\cdots 0 \rightarrow R \xrightarrow{1-x} R \xrightarrow{1-x} R \rightarrow \cdots$$

and we deduce that $(R_x)^{\text{h}G} \simeq k$ in $\text{D}(R)$. But $k \in \text{D}_b(R)$ is not perfect, hence $\chi^G(X) \simeq \text{sing}(k) \neq 0$. Using Proposition 5.14, this means $X \notin \mathcal{P}(G; R)^\natural$.

6. MAIN RESULT

We have established in Theorem 3.21 that the canonical functor (1.2)

$$\bar{F}: \left(\frac{\text{K}_b(\text{perm}(G; R))}{\text{K}_{b, \text{ac}}(\text{perm}(G; R))} \right)^\natural \longrightarrow \text{D}_b(RG)$$

is fully faithful and has essential image the auxiliary category $\mathcal{P}(G; R)^\natural$ of (3.1). We saw in Proposition 5.14 that $\mathcal{P}(G; R)^\natural$ is contained in the subcategory of cohomologically perfect complexes (Definition 5.4). We now want to prove the converse.

Two ideas will be key: the “cohomology” comonad $\text{Infl}_1^G \circ (-)^{\text{h}G}$ on cohomologically perfect objects, and compactness arguments. To make both work at the same time, we lift that comonad to the homotopy category of injectives, $\text{K}(\text{Inj}(RG))$, whose compact part is the bounded derived category. The proof of our main result being pretty long, we prove several shorter lemmas. Let us first set the notation.

6.1. *Recollection.* We are going to construct the following diagram via [Kra05, § 6]

$$(6.2) \quad \begin{array}{ccc} & & \text{K}(\text{Inj}(RG)) \\ & \begin{array}{c} \xrightarrow{Q_\lambda} \\ \xleftarrow{Q} \\ \xrightarrow{Q_\rho} \end{array} & \\ \text{D}(RG) & & \uparrow \hat{c}_* \quad \downarrow \hat{c}! \\ \uparrow c_* \quad \downarrow c! & & \text{K}(\text{Inj}(R)) \\ \text{D}(R) & \begin{array}{c} \xrightarrow{Q_\lambda} \\ \xleftarrow{Q} \\ \xrightarrow{Q_\rho} \end{array} & \end{array}$$

We already encountered the slanted arrows $Q_\lambda \dashv Q \dashv Q_\rho$ in the recollement (4.2). The left-hand vertical arrows $c_* \dashv c!$ are simply a shorthand for (5.2):

$$c_* := \text{Infl}_1^G \quad \text{and} \quad c! := (-)^{\text{h}G}.$$

There are several reasons for this notation. First, it is lighter in formulas involving iterated compositions. Second, it evokes the algebro-geometric notation $c^* \dashv c_* \dashv c!$ for an imaginary closed immersion $c: \text{Spec}(R) \hookrightarrow \text{Spec}(RG)$ – that actually makes sense if G is abelian. (And we do have a left adjoint c^* too, namely the left-derived functor of deflation $(-)^{\text{h}G}$.) Finally, it allows for a simple notation at the level of $\text{K}(\text{Inj})$, namely the yet-to-be-explained $\hat{c}_* \dashv \hat{c}!$ on the right-hand side of (6.2).

For this, we apply [Kra05, §6] to the *exact* functor (denoted F in *loc. cit.*) $\text{Infl}_1^G: \text{Mod}(R) \rightarrow \text{Mod}(RG)$. Its right adjoint $(-)^G: \text{Mod}(RG) \rightarrow \text{Mod}(R)$ preserves injectives and our $\hat{c}^!: \text{K}(\text{Inj}(RG)) \rightarrow \text{K}(\text{Inj}(R))$ is simply $(-)^G$ degreewise. Its left adjoint $\hat{c}_*: \text{K}(\text{Inj}(R)) \rightarrow \text{K}(\text{Inj}(RG))$ is more subtle than just inflation. It is Krause's construction, namely $\hat{c}_*$ is defined as the following composite

$$\hat{c}_*: \text{K}(\text{Inj}(R)) \xrightarrow{J} \text{K}(\text{Mod}(R)) \xrightarrow{\text{Infl}_1^G} \text{K}(\text{Mod}(RG)) \xrightarrow{J_\lambda} \text{K}(\text{Inj}(RG)).$$

where $J: \text{K}(\text{Inj}) \rightarrow \text{K}(\text{Mod})$ is the inclusion and $J_\lambda: \text{K}(\text{Mod}) \rightarrow \text{K}(\text{Inj})$ its left adjoint, as in (4.4). It is an easy exercise to verify (using J fully faithful) that $\hat{c}_* \dashv \hat{c}^!$. (Although we had a derived left adjoint $c^* \dashv c_*$ there is no $\hat{c}^* \dashv \hat{c}_*$ on $\text{K}(\text{Inj})$.)

By [Kra05, Lemma 6.3], since inflation is exact, we have

$$(6.3) \quad Q \circ \hat{c}_* \cong c_* \circ Q: \text{K}(\text{Inj}(R)) \rightarrow \text{D}(RG).$$

From this we deduce, by taking right adjoints, that

$$(6.4) \quad \hat{c}^! \circ Q_\rho \cong Q_\rho \circ c^!.$$

Note that since the functor $(-)^G: \text{Mod}(RG) \rightarrow \text{Mod}(R)$ preserves coproducts, so does the induced $\hat{c}^!$ on $\text{K}(\text{Inj})$. Thus its left adjoint preserves compacts:

$$(6.5) \quad \hat{c}_*(\text{K}(\text{Inj}(R))^c) \subseteq \text{K}(\text{Inj}(RG))^c.$$

6.6. *Remark.* On every $Y \in \text{K}(\text{Inj}(RG))$ the comonad $\hat{c}_*\hat{c}^!$ equals by construction

$$\hat{c}_*\hat{c}^!(Y) = J_\lambda \text{Infl}_1^G J \hat{c}^!(Y) = J_\lambda \text{Infl}_1^G (J(Y)^G) \cong J_\lambda \text{hom}_{RG}(R, Y)$$

where $\text{hom}_{RG}(R, Y)$ has trivial G -action. This leads us to bimodule actions:

6.7. **Lemma.** *There is an action of the bounded derived category of $R(G \times G^{\text{op}})$ -modules on $\text{K}_-(\text{Inj}(RG))$, in the form of a well-defined bi-exact functor*

$$[-, -]: \text{D}_b(R(G \times G^{\text{op}}))^{\text{op}} \times \text{K}_-(\text{Inj}(RG)) \rightarrow \text{K}_-(\text{Inj}(RG))$$

given by the formula $[L, Y] = J_\lambda(\text{hom}_{RG}(L, Y))$.

Proof. On the level of module categories, there is an action

$$\text{hom}_{RG}(-, -): \text{Mod}(R(G \times G^{\text{op}}))^{\text{op}} \times \text{Mod}(RG) \rightarrow \text{Mod}(RG)$$

which takes (L, Y) to the abelian group $\text{Hom}_{RG}(L, Y)$ built by viewing L as a left RG -module via its left G -action, and then making the output $\text{Hom}_{RG}(L, Y)$ into a left G -module $\text{hom}_{RG}(L, Y)$ by using the ‘remaining’ right G -action on L . Being additive in both variables, this passes to homotopy categories

$$(6.8) \quad \text{hom}_{RG}(-, -): \text{K}(R(G \times G^{\text{op}}))^{\text{op}} \times \text{K}(RG) \rightarrow \text{K}(RG)$$

(by totalizing via $\coprod$, which is irrelevant in our bounded case). This yields

$$[-, -]: \text{K}_b(R(G \times G^{\text{op}}))^{\text{op}} \times \text{K}_-(\text{Inj}(RG)) \xrightarrow{(6.8)} \text{K}_-(RG) \xrightarrow{J_\lambda} \text{K}_-(\text{Inj}(RG)).$$

The preservation of left-boundedness by J_λ is Lemma 4.5(b). To show that this descends to the derived category in the first variable, let $L \in \text{K}_b(R(G \times G^{\text{op}}))$ be acyclic and let $Y \in \text{K}_-(\text{Inj}(RG))$, and let us show that $J_\lambda(\text{hom}_{RG}(L, Y)) = 0$. By Lemma 4.5(b) again, it suffices to show that $\text{hom}_{RG}(L, Y)$ is acyclic. But $H_n \text{hom}_{RG}(L, Y) = \text{Hom}_{\text{K}(RG)}(L[n], Y)$ vanishes since Y is a left-bounded complex of injectives and L acyclic (as complex of RG -modules as well). $\square$

6.9. *Remark.* Each object L in $D_b(R(G \times G^{\text{op}}))$ thus defines an exact endofunctor

$$[L, -]: K_-(\text{Inj}(RG)) \rightarrow K_-(\text{Inj}(RG)).$$

For instance, $[RG, -] \cong \text{Id}$ whereas $[R, -] \cong \hat{c}_* \hat{c}^!$ is our comonad, by Remark 6.6. We use this to decide when $Y \in K_-(\text{Inj}(RG))$ can be recovered from $\hat{c}_* \hat{c}^!(Y)$.

6.10. **Lemma.** *Let G be a finite p -group and $Y \in K_-(\text{Inj}(RG))$, left-bounded and such that $p^n \cdot \text{id}_Y = 0$ for $n \gg 1$. Then Y belongs to $\text{thick}(\hat{c}_* \hat{c}^!(Y))$ in $K_-(\text{Inj}(RG))$.*

Proof. As explained in Remark 6.9, we need to show that in $K_-(\text{Inj}(RG))$

$$[RG, Y] \in \text{thick}([R, Y]).$$

Since $p^n \cdot Y = 0$, we also have $p^n \cdot [RG, Y] = 0$ and Remark 3.7 gives

$$[RG, Y] \in \text{thick}(\text{cone}([RG, Y] \xrightarrow{p} [RG, Y])) = \text{thick}([\text{cone}(RG \xrightarrow{p} RG), Y])$$

using exactness of $[-, Y]$. Hence it suffices to prove in $D_b(R(G \times G^{\text{op}}))$ that

$$\text{cone}(RG \xrightarrow{p} RG) \in \text{thick}(R).$$

By Proposition 2.24 for $\mathbb{Z} \rightarrow R$, it suffices to prove that in $D_b(\mathbb{Z}(G \times G^{\text{op}}))$

$$\text{cone}(\mathbb{Z}G \xrightarrow{p} \mathbb{Z}G) \in \text{thick}(\mathbb{Z}).$$

Consider the exact functor $i_*: D_b(\mathbb{F}_p(G \times G^{\text{op}})) \rightarrow D_b(\mathbb{Z}(G \times G^{\text{op}}))$. The above $\text{cone}(\mathbb{Z}G \xrightarrow{p} \mathbb{Z}G)$ is nothing but $i_*(\mathbb{F}_p G)$ and $i_*(\mathbb{F}_p) \cong \text{cone}(\mathbb{Z} \xrightarrow{p} \mathbb{Z})$ belongs to $\text{thick}(\mathbb{Z})$. So we are reduced to show that $\mathbb{F}_p G \in \text{thick}(\mathbb{F}_p)$ in $D_b(\mathbb{F}_p(G \times G^{\text{op}}))$, which is clear since $D_b(\mathbb{F}_p(G \times G^{\text{op}})) = \text{thick}(\mathbb{F}_p)$ as $G \times G^{\text{op}}$ is a also p -group. $\square$

6.11. **Lemma.** *Let G be a p -group. Let $X \in D_b(RG)$ be p -torsion (Proposition 3.5) and G -cohomologically perfect (Definition 5.4). Then X belongs to $\text{thick}(c_* R)$.*

Proof. By Lemma 4.7, the assumption $0 = \chi^G(X) = I_\lambda Q_\rho c^!(X)$ implies that

$$(6.12) \quad Q_\rho c^!(X) \in \text{Loc}(Q_\rho(R))$$

in $K(\text{Inj}(R))$. Applying to this relation the (coproduct-preserving) left adjoint $\hat{c}_*: K(\text{Inj}(R)) \rightarrow K(\text{Inj}(RG))$ of (6.2), we obtain in $K(\text{Inj}(RG))$

$$(6.4) \quad \hat{c}_* \hat{c}^! Q_\rho(X) \cong \hat{c}_* Q_\rho c^!(X) \underset{(6.12)}{\in} \text{Loc}(\hat{c}_* Q_\rho(R)).$$

Hence by Lemma 6.10 with $Y = Q_\rho(X)$, which is p -torsion since X is, we have

$$(6.13) \quad Q_\rho(X) \in \text{thick}(\hat{c}_* \hat{c}^! Q_\rho(X)) \subseteq \text{Loc}(\hat{c}_* Q_\rho(R))$$

in $K(\text{Inj}(RG))$. Now $Q_\rho(X)$ is compact in $K(\text{Inj}(RG))$ by Recollection 4.1 and $\hat{c}_* Q_\rho(R)$ is compact because $Q_\rho(R)$ is and $\hat{c}_*$ preserves compacts (6.5). So, by [Nee92, Lemma 2.2], we can replace ‘Loc’ by ‘thick’ in (6.13), giving us the relation

$$Q_\rho(X) \in \text{thick}(\hat{c}_* Q_\rho(R))$$

in $K(\text{Inj}(RG))$. Applying $Q: K(\text{Inj}(RG)) \rightarrow D(RG)$ and $QQ_\rho \cong \text{Id}$, we get

$$X \in \text{thick}(Q\hat{c}_* Q_\rho(R)) \stackrel{(6.3)}{=} \text{thick}(c_* QQ_\rho(R)) = \text{thick}(c_*(R)). \quad \square$$

6.14. **Lemma.** *Let G be a p -group and $X \in D_b(RG)$. The following are equivalent:*

- (i) $X \in \mathcal{P}(G; R)^\natural$; see (3.1).
- (ii) X is cohomologically perfect (Definition 5.4).
- (iii) X is G -cohomologically perfect and R -perfect (Definition 2.19).

Proof. The implication (i)⇒(ii) is Proposition 5.14, and the implication (ii)⇒(iii) is trivial by Definition 5.4. For the implication (iii)⇒(i) suppose that $\chi^G(X) = 0$ and X is R -perfect. By Corollary 3.6, there exists an exact triangle in $D_b(RG)$

$$P \rightarrow X \oplus \Sigma X \rightarrow T \rightarrow \Sigma P$$

where P is a bounded complex of permutation modules (hence belongs to $\mathcal{P}(G; R)^\natural$ by Corollary 3.17) and where $T \in D_b(RG)$ is p -torsion. Note that T remains G -cohomologically perfect and R -perfect since P and X are. In other words, we reduce to the case where X is p -torsion. But then Lemma 6.11 tells us that $X \in \text{thick}(c_*R)$. We conclude by Corollary 3.17 which guarantees that $c_*(R) \in \mathcal{P}(G; R)^\natural$. $\square$

6.15. *Remark.* For G a p -group the equivalence (ii)⇔(iii) in Lemma 6.14 shows that G -cohomological perfection together with R -perfection does imply H -cohomological perfection for all $H \leq G$. This is sharp by Example 5.9 and Example 5.16.

Here is the main result. The functor $\bar{F}$ is in (1.2). The subcategory $\mathcal{P}(G; R)^\natural \subseteq D_b(RG)$ is in (3.1). The invariant χ^H is in Definition 5.4. See also Theorem 3.21.

6.16. **Theorem.** *Let G be a finite group and R a commutative noetherian ring. Let $X \in D_b(RG)$ be a bounded complex. The following properties of X are equivalent:*

- (i) *The complex X belongs to $\text{Im}(\bar{F}) = \mathcal{P}(G; R)^\natural$.*
- (ii) *It is cohomologically perfect: $\chi^H(X) = 0$ in $D^{\text{sing}}(R)$ for all subgroups $H \leq G$.*
- (iii) *It is R -perfect, i.e. the underlying complex $\text{Res}_1^G(X) \in D_b(R)$ is perfect, and it is H -cohomologically perfect, $\chi^H(X) = 0$, for every Sylow subgroup $H \leq G$.*
- (iv) *It is E -cohomologically perfect for every elementary abelian subgroup $E \leq G$.*
- (v) *In $D(R)$, we have $X^{hH} \in \text{thick}(\{R^{hK} \mid K \leq G\})$ for every subgroup $H \leq G$.*

Proof. Implication (i)⇒(ii) is Proposition 5.14. The implication (ii)⇒(iii) is trivial (Definition 5.4). The equivalence between (iii) and (iv) follows from Lemma 6.14, as explained in Remark 6.15. On the other hand, assuming (iii), Lemma 6.14 tells us that $\text{Res}_H^G(X) \in \mathcal{P}(H; R)^\natural$ for every Sylow subgroup $H \leq G$. This implies (i) by Corollary 3.19. So the four conditions (i)⇔(ii)⇔(iii)⇔(iv) are equivalent.

Let us denote by $\mathcal{J} := \text{thick}(\{R^{hK} \mid K \leq G\})$ the thick subcategory of $D(R)$ that appears in (v). For (v)⇒(ii), note that $\text{sing}(R^{hK}) = \chi^K(R) = 0$ by Corollary 5.15. So $\mathcal{J} \subseteq \text{Ker}(\text{sing})$ and therefore $X^{hH} \in \mathcal{J}$ implies $\chi^H(X) = \text{sing}(X^{hH}) = 0$. Finally, for (i)⇒(v), since $\mathcal{P}(G; R)^\natural = \text{Im}(\bar{F}) = \text{thick}(\text{perm}(G; R))$ by Theorem 3.21, it suffices to prove that for every subgroups $H, L \leq G$ we have $(R(G/L))^{hH} \in \mathcal{J}$. This follows from the Mackey formula, Proposition 5.10 and the definition of $\mathcal{J}$. $\square$

6.17. *Remark.* As in Remark 3.20, we can use Corollary 2.20 to only test (iii), or (iv), for those p -subgroups of G with p non-invertible on X (and in R).

6.18. *Remark.* Theorem 1.4 of the Introduction follows from Theorems 3.21 and 6.16.

6.19. *Remark.* The inflation functor $c_* : D(R) \rightarrow D(RG)$ is monoidal and its right adjoint $c^! = (-)^{hG} : D(RG) \rightarrow D(R)$ is therefore lax monoidal. In particular, $c^!c_*(\mathbb{1}) = R^{hG}$ is a ring object, namely the ‘cohomology ring’ of G with coefficients in R , and every object $X \in D(RG)$ gives rise to a module X^{hG} over this ring.

With this in mind, and the fact that for every ring Λ we have $D_{\text{perf}}(\Lambda) = \text{thick}(\Lambda)$, the terminology ‘cohomologically perfect’ of Definition 5.4 is somewhat justified by the equivalent formulation given in part (v) of Theorem 6.16.

The regular case is now trivial (compare Corollary 1.7 and Scholium 3.15):

6.20. Corollary. *Suppose that R is regular. Then $\mathcal{P}(G; R)^{\natural} = D_b(RG)$ and the canonical functor $K_b(RG) \rightarrow D_b(RG)$ induces an equivalence*

$$(6.21) \quad \left(\frac{K_b(\text{perm}(G; R))}{K_{b,ac}(\text{perm}(G; R))} \right)^{\natural} \xrightarrow{\sim} D_b(RG).$$

Proof. We have $\mathcal{P}(G; R)^{\natural} = D_b(RG)$ by Theorem 6.16 since all obstructions $\chi^H(X)$ trivially vanish in $D^{\text{sing}}(R) = 0$. We conclude by Theorem 3.21. $\square$

7. LOCALIZATION OF BIG CATEGORIES

Neeman's Theorem [Nee92] suggests that the equivalence (6.21) in Corollary 6.20 might be the compact tip of an iceberg, which indeed emerges in this section.

7.1. Definition. The big homotopy category $K(\text{Mod}(RG))$ of RG -modules admits small coproducts. We define the (*big*) *derived category of permutation modules*

$$D\text{Perm}(G; R) := \text{Loc}(\text{perm}(G; R)) = \text{Loc}(\{ R(G/H) \mid H \leq G \})$$

as the localizing subcategory of $K(\text{Mod}(RG))$ generated by permutation modules.

Since every generator $R(G/H)$ is compact in $K(\text{Mod}(RG))$, we know by [Nee92] that the triangulated category $D\text{Perm}(G; R)$ is compactly generated and its subcategory of compact objects is the bounded homotopy category of $\natural$ -permutation modules (see Recollection 1.11):

$$(7.2) \quad D\text{Perm}(G; R)^c = \text{thick}(\text{perm}(G; R)) = K_b(\text{perm}(G; R)^{\natural}).$$

7.3. Remark. This *ad hoc* definition does not do justice to the derived category of permutation modules but it has the advantage of simplicity and catches the right compacts. Let us now give a more conceptual approach, justifying the name.

7.4. Notation. Let $\text{Perm}(G; R) \subseteq \text{Mod}(RG)$ be the category of all permutation RG -modules $R(A)$ for possibly infinite G -sets A . It is the closure of $\text{perm}(G; R)$ under coproducts. We have inclusions $D\text{Perm}(G; R) \subseteq K(\text{Perm}(G; R)) \subseteq K(\text{Mod}(RG))$. Define the class QI_G of G -quasi-isomorphisms in $K(\text{Perm}(G; R))$ by

$$QI_G = \left\{ s : X \rightarrow Y \text{ in } K(\text{Perm}(G; R)) \mid \begin{array}{l} s^H : X^H \rightarrow Y^H \text{ is a} \\ \text{quasi-isomorphism for all } H \leq G \end{array} \right\}.$$

7.5. Proposition. *The category $D\text{Perm}(G; R)$ is obtained from the homotopy category of all permutation modules by inverting G -quasi-isomorphisms: The composite $D\text{Perm}(G; R) \hookrightarrow K(\text{Perm}(G; R)) \twoheadrightarrow K(\text{Perm}(G; R))[QI_G^{-1}]$ is an equivalence.*

Proof. Since $\mathcal{D} := D\text{Perm}(G; R)$ is generated by the compacts $R(G/H)$ of $\mathcal{K} := K(\text{Perm}(G; R))$, it follows by Brown-Neeman Representability that the inclusion $\mathcal{D} \hookrightarrow \mathcal{K}$ admits a right adjoint and the composite $\mathcal{D} \hookrightarrow \mathcal{K} \twoheadrightarrow \mathcal{K}/\mathcal{D}^{\perp}$ is an equivalence. Here we have $\mathcal{D}^{\perp} = \{ \text{cone}(s) \mid s \in QI_G \}$ since for all $Z \in \mathcal{K}$, $H \leq G$ and $n \in \mathbb{Z}$

$$\text{Hom}_{\mathcal{K}}(R(G/H)[n], Z) \cong \text{Hom}_{K(\text{Mod}(R))}(R[n], Z^H) \cong H_n(Z^H). \quad \square$$

7.6. Remark. The above quotient functor $\mathcal{K} \twoheadrightarrow \mathcal{K}[QI_G^{-1}]$ realizes $D\text{Perm}(G; R)$ as a Bousfield colocalization of $\mathcal{K} = K(\text{Perm}(G; R))$. It is also a Bousfield localization (i.e. that quotient also admits a fully faithful right adjoint) because $K(\text{Perm}(G; R))$ is itself compactly generated. See details in the expository paper [BG20].

7.7. *Remark.* In [BG20] we show that $\mathrm{DPerm}(G; R)$ is equivalent to the derived category of cohomological R -linear Mackey functors on G . After suitably extending Definition 7.1 to profinite groups, we also explain in [BG20] how $\mathrm{DPerm}(G; R)$ is equivalent to the triangulated category of Artin motives $\mathrm{DAM}(\mathbb{F}; R)$ in the sense of Voevodsky, over a base field $\mathbb{F}$ whose absolute Galois group is G .

Let us now return to the relation with $\mathrm{K}(\mathrm{Inj}(RG))$.

7.8. *Definition.* Define a localizing subcategory of $\mathrm{K}(\mathrm{Inj}(RG))$ as follows:

$$\mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R) := \mathrm{Loc} (Q_\rho(R(G/H)) \mid H \leq G).$$

Since the functor $Q_\rho : \mathrm{D}(RG) \rightarrow \mathrm{K}(\mathrm{Inj}(RG))$ of (4.2) identifies $\mathrm{D}_b(\mathrm{mod}(G; R))$ with the compact objects of $\mathrm{K}(\mathrm{Inj}(RG))$, we see that $\mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R)$ is a compactly generated triangulated category, and its compact part is equivalent to $\mathcal{P}(G; R)^\natural$:

$$(7.9) \quad \mathcal{P}(G; R)^\natural = \mathrm{thick}(R(G/H) \mid H \leq G) \xrightarrow[\cong]{Q_\rho} \mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R)^c.$$

The two subcategories we defined so far are related by the functor J_λ of (4.4).

7.10. **Lemma.** *For every subgroup $H \leq G$, we have $J_\lambda(R(G/H)) \cong Q_\rho(R(G/H))$ in $\mathrm{K}(\mathrm{Inj}(RG))$. Consequently, $J_\lambda(\mathrm{DPerm}(G; R)) \subseteq \mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R)$.*

Proof. The first claim follows from Lemma 4.5 (c) since $R(G/H)$ is clearly bounded. The second claim follows by Definitions 7.1 and 7.8 and cocontinuity of J_λ . $\square$

7.11. *Notation.* In view of Lemma 7.10, the restriction of J_λ to $\mathrm{DPerm}(G; R)$ yields a well-defined coproduct-preserving and compact-preserving exact functor

$$(7.12) \quad F^+ := (J_\lambda)|_{\mathrm{DPerm}(G; R)} : \mathrm{DPerm}(G; R) \rightarrow \mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R).$$

This functor F^+ extends beyond compacts the functor F of the Introduction:

7.13. **Proposition.** *The following diagram commutes up to isomorphism*

$$(7.14) \quad \begin{array}{ccccc} \mathrm{K}_b(\mathrm{perm}(G; R))^\natural & \xlongequal{\quad} & \mathrm{DPerm}(G; R)^c & \xrightarrow{\quad} & \mathrm{DPerm}(G; R) \\ \text{(see Cor. 3.17)} \downarrow F & & (F^+)^c \downarrow & & F^+ \downarrow \text{(see (7.12))} \\ \mathcal{P}(G; R)^\natural & \xrightarrow[\cong]{(7.9)} & \mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R)^c & \xrightarrow{\quad} & \mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R) \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{D}_b(\mathrm{mod}(RG)) & \xrightarrow[\cong]{Q_\rho} & \mathrm{K}(\mathrm{Inj}(RG))^c & \xrightarrow{\quad} & \mathrm{K}(\mathrm{Inj}(RG)). \end{array}$$

Proof. Simply Lemma 4.5 (c) again, since $\mathrm{K}_b(\mathrm{perm}(G; R))^\natural \subseteq \mathrm{K}_-(\mathrm{Mod}(RG))$. $\square$

7.15. **Theorem.** *Let G be a finite group and R a commutative noetherian ring. The functor $F^+ : \mathrm{DPerm}(G; R) \rightarrow \mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R)$ is a finite localization with respect to $\mathrm{K}_{b, \mathrm{ac}}(\mathrm{perm}(G; R))$, i.e. it induces a well-defined equivalence*

$$(7.16) \quad \bar{F}^+ : \frac{\mathrm{DPerm}(G; R)}{\mathrm{Loc}(\mathrm{K}_{b, \mathrm{ac}}(\mathrm{perm}(G; R)))} \xrightarrow{\sim} \mathrm{K} \mathrm{Inj}_{\mathrm{perm}}(G; R).$$

On compacts, this equivalence identifies with the equivalence $\bar{F}$ of (3.22), via (7.9).

Proof. By Lemma 4.5 (b), for every bounded acyclic $X \in \mathrm{K}_{b, \mathrm{ac}}(\mathrm{perm}(G; R))$ we have $F^+(X) = J_\lambda(X) = 0$. We deduce that F^+ descends to the quotient as in (7.16). This functor $\bar{F}^+$ preserves coproducts and compact objects hence it suffices to show it is an equivalence on compact objects. As $\bar{F}^+$ agrees with $\bar{F}$ on compacts by Proposition 7.13, this equivalence on compacts holds by Theorem 3.21. $\square$

7.17. Corollary. *If R is regular then we have $\mathrm{KInj}_{\mathrm{perm}}(G; R) = \mathrm{K}(\mathrm{Inj}(RG))$ and the functor $F^+ : \mathrm{DPerm}(G; R) \rightarrow \mathrm{K}(\mathrm{Inj}(RG))$ is a finite localization with respect to $\mathrm{K}_{\mathrm{b}, \mathrm{ac}}(\mathrm{perm}(G; R))$. On compacts, F^+ induces the localization of Corollary 6.20.*

Proof. This follows from Theorem 7.15, using Corollary 6.20. $\square$

7.18. Remark. Assume that $R = k$ is a field. We mention as a curiosity that, as localizing subcategories of $\mathrm{K}(\mathrm{Mod}(RG))$, we have the inclusion

$$(7.19) \quad \mathrm{K}(\mathrm{Inj}(kG)) \subseteq \mathrm{DPerm}(G; k)$$

and the finite localization F^+ is simply the left adjoint to that inclusion. This is quite remarkable: For example, it says that for each finitely generated kG -module M with injective resolution I_M , the latter may be constructed out of permutation modules using triangles, suspensions and (arbitrary) coproducts. In fact, these I_M are generators for $\mathrm{K}(\mathrm{Inj}(kG))$ so this fact is equivalent to the inclusion (7.19).

To prove it, we may assume the k -linear dual M^* belongs to $\mathcal{P}(G; k)$ and choose a 1-free permutation resolution $P(1) \rightarrow M^*$. As $P(1) \in \mathcal{P}(G; k)$ by Proposition 2.7, we can find a 2-free permutation resolution $P(2) \rightarrow P(1)$. Continuing like this and passing to k -linear duals we construct a sequence

$$(7.20) \quad I(1) \xrightarrow{f(1)} I(2) \xrightarrow{f(2)} I(3) \xrightarrow{f(3)} \dots$$

where each $I(n)$ is a bounded complex of permutation modules with $I(n)_i$ injective for all $i \geq -n$, and each $f(n)$ is a quasi-isomorphism. Note that the filtered colimit $I_M = \mathrm{colim}_n I(n)$ is a resolution of M , and in each degree i , the sequence

$$(7.21) \quad I(1)_i \rightarrow I(2)_i \rightarrow \dots$$

eventually consists of injective kG -modules. It follows that I_M is an injective resolution of M . (Indeed, injective, projective and flat modules coincide, and filtered colimits of flat are flat.) The final two steps are to show that $\mathrm{Perm}(G; k)$ is closed under filtered colimits in $\mathrm{Mod}(kG)$, which follows from the comparison with cohomological Mackey functors in [BG20], and that $\mathrm{DPerm}(G; k)$ is closed under sequential colimits in $\mathrm{Ch}(\mathrm{Perm}(G; k))$, analogously to [BK08, Theorem 13.3].

8. DENSITY AND GROTHENDIECK GROUP

We want to use Thomason's classification of dense subcategories to derive consequences from the results of Sections 2 and 3. Although some generalities work for any *regular* ring R , we are primarily interested in the field case.

8.1. Hypothesis. In this section, k is a field of characteristic $p > 0$.

8.2. Recollection. Given an essentially small triangulated category $\mathcal{T}$ we may consider its Grothendieck group, $\mathrm{K}_0(\mathcal{T})$, the free abelian group generated by isomorphism classes of objects in $\mathcal{T}$ quotiented by the relation $[X] + [Z] = [Y]$ for each exact triangle $X \rightarrow Y \rightarrow Z \rightarrow \Sigma X$ in $\mathcal{T}$. In particular $-[X] = [\Sigma X]$.

For each dense triangulated subcategory $\mathcal{A} \subseteq \mathcal{T}$ (Recollection 2.11) the map $\mathrm{K}_0(\mathcal{A}) \rightarrow \mathrm{K}_0(\mathcal{T})$ is injective ([Tho97, Corollary 2.3]) hence $\mathrm{K}_0(\mathcal{A})$ defines a subgroup of $\mathrm{K}_0(\mathcal{T})$. Conversely, each subgroup $L \subseteq \mathrm{K}_0(\mathcal{T})$ defines a dense subcategory

$$\mathcal{A}(L) := \{ X \in \mathcal{T} \mid [X] \in L \text{ in } \mathrm{K}_0(\mathcal{T}) \}.$$

By [Tho97, Theorem 2.1] these constructions yield a well-defined bijection

$$\{ \text{dense triangulated subcategories of } \mathcal{T} \} \xrightarrow{\sim} \{ \text{subgroups of } \mathrm{K}_0(\mathcal{T}) \}.$$

8.3. *Remark.* We want to apply Thomason's Theorem to the triangulated subcategories $\mathcal{P}(G; k)$ and $\mathcal{Q}(G; k)$ of $\mathcal{T} = D_b(kG)$ introduced in Definition 2.8, which are dense by Corollary 3.11. The Grothendieck group of $D_b(kG)$ as a triangulated category coincides with the Grothendieck group of $\text{mod}(kG)$ as an abelian category

$$K_0(D_b(\text{mod}(kG))) \cong K_0(\text{mod}(kG)) = G_0(kG).$$

This Grothendieck group is free abelian on the set of isomorphism classes of simple kG -modules. In the classic reference [Ser77], Serre writes $R_k(G)$ for $G_0(kG)$ and $P_k(G)$ for the Grothendieck group $K_0(kG)$ of projective modules. We do *not* adopt this notation to avoid confusion with our coefficient ring R and the category $\mathcal{P}(G; k)$.

Let us gather in one statement all the subgroups of $G_0(kG)$ that we consider.

8.4. **Proposition.** *We have the inclusions of subgroups in $G_0(kG) = K_0(\text{mod}(kG))$*

$$\begin{array}{ccc} \mathbb{Z} \cdot [kG] & \subseteq & K_0(kG) = K_0(\text{proj}(kG)) \\ \uparrow \cap & & \uparrow \cap \\ K_0^{\mathcal{P}}(G; k) := K_0(\mathcal{P}(G; k)) & \subseteq & K_0^{\mathcal{Q}}(G; k) := K_0(\mathcal{Q}(G; k)) \subseteq G_0(kG). \end{array}$$

The quotients $G_0(kG)/K_0(kG)$ and $G_0(kG)/K_0^{\mathcal{Q}}(G; k)$ are finite. Specifically, these quotients are finite abelian p -groups, whose exponent is a power of p dividing $|G|$.

Proof. Injectivity of $K_0^{\mathcal{P}}(G; k) \rightarrow G_0(kG)$ and $K_0^{\mathcal{Q}}(G; k) \rightarrow G_0(kG)$ follows from density (Corollary 3.11) and Recollection 8.2. All inclusions in the statement are then straightforward, already for underlying categories. The finite index of the subgroup $K_0(kG)$, and therefore of $K_0^{\mathcal{Q}}(G; k)$, follows from [Ser77, § 16, Theorem 35]: There exists a power p^n dividing $|G|$ such that $p^n \cdot G_0(kG) \subseteq K_0(kG)$. ⁽¹⁾ $\square$

8.5. *Example.* The cokernel of the ‘Cartan’ homomorphism $K_0(RG) \rightarrow G_0(RG)$ is not always of finite exponent when R is not a field, even for a DVR. Take $R = \mathbb{Z}_{(2)}$ and $G = C_2$ cyclic of order 2. Then $K_0(RC_2) = \mathbb{Z} \cdot [RC_2]$ since RC_2 is local. Let $R^+ = R$ with trivial C_2 -action. Rationally, in $G_0(\mathbb{Q}C_2) \cong \mathbb{Z} \cdot [\mathbb{Q}^+] \oplus \mathbb{Z} \cdot [\mathbb{Q}^-]$ for $\mathbb{Q}^+$ trivial and $\mathbb{Q}^- = \mathbb{Q}$ with sign action, our $[R^+]$ maps to $[\mathbb{Q}^+]$ but $[RC_2]$ maps to $[\mathbb{Q}^+] + [\mathbb{Q}^-]$. So no non-zero multiple of $[R^+] \in G_0(RC_2)$ belongs to $K_0(RC_2)$.

Applying Thomason's classification (Recollection 8.2) we get for instance:

8.6. **Corollary.** *An $M \in \text{mod}(kG)$ admits m -free permutation resolutions for all $m \geq 0$ if and only if its class $[M] \in G_0(kG)$ belongs to the subgroup $K_0^{\mathcal{P}}(G; k)$. (Same for m -projective p -permutation resolutions, with $K_0^{\mathcal{Q}}(G; k)$ instead.) $\square$*

We do not have a description of $K_0^{\mathcal{P}}(G; k)$ in general but it is already remarkable to have a condition in terms of the class of M in the Grothendieck group. Using only that free modules belong to $\mathcal{P}(G; k)$ we get some interesting consequences.

8.7. **Corollary.** *Let $M \in \text{mod}(kG)$ and consider $\Omega(M) = \text{Ker}(kG \otimes_k M \rightarrow M)$. Then $M \oplus \Omega(M)$ admits m -free permutation resolutions for all $m \geq 0$. In particular, $M \oplus \Omega(M)$ admits a finite resolution by finitely generated permutation modules.*

Proof. We have in $G_0(kG)$ that $[M \oplus \Omega(M)] = [kG \otimes_k M] \in K_0^{\mathcal{P}}(G; k)$ since $kG \otimes_k M$ is free. So we apply Corollary 8.6. The second part follows by Corollary 2.5. $\square$

¹ The general assumptions of [Ser77, p. 115] hold for any k by [Hoc17, Theorem, p. 23].

8.8. *Remark.* Unlike $K_0^Q(G; k)$, the subgroup $K_0^P(G; k) \subseteq G_0(kG)$ is not of finite index in general, simply because permutation modules are defined integrally. So the subgroup $K_0^P(G; k)$ is always contained in the image of $G_0(\mathbb{F}_p G)$ inside $G_0(kG)$ that can have infinite index when kG has simple modules not defined over $\mathbb{F}_p$.

8.9. *Remark.* The essential images of the canonical functors $K_b(\text{perm}(G; k)) \rightarrow D_b(kG)$ and $K_b(\text{perm}(G; k)^\natural) \rightarrow D_b(kG)$ are also dense triangulated subcategories. Indeed, these essential images are the same as those of the functors

$$\frac{K_b(\text{perm}(G; k))}{K_{b,ac}(\text{perm}(G; k))} \rightarrow D_b(kG) \quad \text{and} \quad \frac{K_b(\text{perm}(G; k)^\natural)}{K_{b,ac}(\text{perm}(G; k)^\natural)} \rightarrow D_b(kG).$$

As these functors are full by Theorem 3.21, their images are triangulated subcategories. As these images contain $\mathcal{P}(G; k)$, they are dense by Corollary 3.11. In fact, the right-hand functor is already essentially surjective, as we shall see in Theorem 8.17. By Thomason, it suffices to understand what happens on K_0 .

8.10. **Proposition** (Boltje/Bouc). *The canonical homomorphism*

$$K_0(\text{perm}(G; k)^\natural) \rightarrow G_0(kG)$$

from the additive Grothendieck group of p -permutation modules (a. k. a. the p -permutation ring [BT10], or trivial source ring [Bol98]) is a surjection onto $G_0(kG)$.⁽²⁾

Proof. Brauer's Theorem in the modular case [Ser77, § 17.2] asserts that

$$\text{Ind} : \bigoplus_H G_0(kH) \twoheadrightarrow G_0(kG)$$

is surjective, where H runs through the so-called Γ_K -elementary subgroups of G (with notation of [Ser77, § 12.4]). So it suffices to prove the result for G of that type. In that case, we prove that every *simple* kG -module M is p -permutation.

In the ‘easy case’ where $G = C \rtimes Q$ with C (cyclic) of order a power of p and Q of order prime to p , we can consider the non-zero submodule M^C of M . As C is normal in G , it follows that M^C is a kG -submodule of M , hence equal to it. In short, M has trivial restriction to the p -Sylow C of G , hence is p -permutation.

The ‘tricky case’ is when $G = C \rtimes P$ where P is a p -Sylow and C is cyclic of order m prime to p . Using induction on $|G|$ as in the proof of [Ser77, § 17.3, Theorem 41], we reduce to the case where M is a finite extension $k' = k[X]/f(X)$ where f is an irreducible factor of $X^m - 1$, on which P acts through k -automorphisms of the field k' . As m and p are coprime, the cyclotomic extension k'/k is separable and hence Galois. It follows from the normal basis theorem that the $k[P]$ -module k' is permutation, and we conclude as before (see Recollection 1.11). $\square$

8.11. *Remark.* If we assume the field k ‘sufficiently large’ (cf. [Ser77, p. 115]), e.g. algebraically closed, the above Γ_K -elementary subgroups are q -elementary for a prime q , i.e. of the form $C \times Q$ for a q -group Q and a cyclic group C of order prime to q . In that case, the p -Sylow of G is normal and we can apply the ‘easy case’ of the above proof. The ‘tricky case’ was communicated to us by Serge Bouc.

8.12. *Remark.* Robert Boltje gave us a different argument to remove ‘ k sufficiently large’ in Proposition 8.10, building a natural section of $K_0(\text{perm}(G; k)^\natural) \rightarrow G_0(kG)$. This uses the canonical induction formula for the Brauer character ring, as well as Galois descent to reduce to the case of k sufficiently large discussed above.

² This homomorphism is rarely injective: Already $G = C_p$, we get $\mathbb{Z}[x]/(x^2 - px) \rightarrow \mathbb{Z}$.

In the remainder of this section, we are going to prove that in fact $\mathcal{Q}(G; k) = D_b(kG)$. Let us start with an easy observation ([Dre75, Theorem 1]).

8.13. Lemma. *Let M be a finitely generated kG -module that admits a k -basis B with the property that $G \cdot B \subseteq B \cup (-B)$, that is, for every $g \in G$ and $b \in B$ we have $gb \in B$ or $-gb \in B$. Then M is a p -permutation module.*

Proof. If $\text{char}(k) = p = 2$ then M is permutation. So assume p odd. It suffices to show that M restricts to a permutation module over a p -Sylow. So we can assume that G is a p -group and show in that case that M is also permutation.

Let $m = \dim_k(M)$. Embed the elementary abelian 2-group $(C_2)^m \hookrightarrow \text{Gl}_m(k)$ as $(\pm 1)^m$ diagonally and the symmetric group $\mathfrak{S}_m \hookrightarrow \text{Gl}_m(k)$ as permutation matrices and let $\Gamma = (C_2)^m \cdot \mathfrak{S}_m \leq \text{Gl}_m(k)$ the subgroup of matrices that have exactly one ± 1 entry in each row and each column and all other entries zero. One shows easily that $\Gamma = (C_2)^m \rtimes \mathfrak{S}_m$ and in any case $[\Gamma : \mathfrak{S}_m] = 2^m$. The basis B in the statement yields a group homomorphism $f : G \rightarrow \text{Gl}_m(k)$ that lands inside Γ by hypothesis.

Now $f(G)$ is a p -subgroup of Γ , hence contained in a p -Sylow. Since $[\Gamma : \mathfrak{S}_m]$ is prime to p , we can assume that up to Γ -conjugation $f(G)$ is contained in $\mathfrak{S}_m$. This means that up to reordering the basis and changing some signs, we can assume that G acts on B via the action of $\mathfrak{S}_m$ on k^m , that is, by permuting the basis. $\square$

8.14. Example. Let A be a finite G -set with n elements and let $V = k(A)$ the corresponding permutation module. Let $0 \leq s \leq n$ and $\Lambda^s V$ the s -th exterior power of V over k . It has dimension $\binom{n}{s}$ and it inherits a ‘diagonal’ G -action as a quotient of $V^{\otimes s}$, that is, such that $g \cdot (v_1 \wedge \cdots \wedge v_s) = (gv_1) \wedge \cdots \wedge (gv_s)$.

Now if we order the elements of A as $A = \{a_1, \dots, a_n\}$ then $\Lambda^s V$ has the usual basis $B = \{a_{i_1} \wedge \cdots \wedge a_{i_s} \mid 1 \leq i_1 < \cdots < i_s \leq n\}$. This basis satisfies the assumptions of Lemma 8.13. Indeed $(ga_{i_1}) \wedge \cdots \wedge (ga_{i_s})$ is equal to an element of B up to the sign of the permutation of s letters that rearranges $ga_{i_1}, \dots, ga_{i_s} \in A$ in increasing order. Hence $\Lambda^s V$ is a p -permutation module.

Choose an order on the elements of $G = \{g_1, \dots, g_n\}$.

8.15. Recollection. (We follow the conventions of [Sta20, Tag 0621] for Koszul complexes.) Let $\text{Kos}(G; k)$ be the Koszul complex of k -vector spaces associated with the augmentation morphism $\epsilon : kG \rightarrow k$ considered as a morphism of k -vector spaces $k^{\oplus n} \rightarrow k$ which is the identity on each summand. Thus explicitly, $\text{Kos}(G; k)$ is concentrated in degrees $n, \dots, 0$, and in degree s is the exterior product $\Lambda_k^s(kG)$ as in Example 8.14 for $A = G$. Moreover, the differential $\text{Kos}(G; k)_s \rightarrow \text{Kos}(G; k)_{s-1}$ sends a basis element $g_{i_1} \wedge \cdots \wedge g_{i_s}$ to the alternating sum

$$\sum_{j=1}^s (-1)^{j-1} g_{i_1} \wedge \cdots \wedge \widehat{g_{i_j}} \wedge \cdots \wedge g_{i_s}$$

where the factor $\widehat{g_{i_j}}$ is removed.

8.16. Lemma. *With the canonical ‘diagonal’ G -action in each degree (Example 8.14), the complex $\text{Kos}(G; k)$ is an acyclic complex of kG -modules.*

Proof. Acyclic is standard (over k), see for instance [Sta20, Tag 0663]. For G -linearity of the differentials we compute for $g \in G$:

$$g \cdot \sum_{j=1}^s (-1)^{j-1} g_{i_1} \wedge \cdots \wedge \widehat{g_{i_j}} \wedge \cdots \wedge g_{i_s} = \sum_{j=1}^s (-1)^{j-1} gg_{i_1} \wedge \cdots \wedge \widehat{gg_{i_j}} \wedge \cdots \wedge gg_{i_s}$$

which is the image of $gg_{i_1} \wedge \cdots \wedge gg_{i_s} = g(g_{i_1} \wedge \cdots \wedge g_{i_s})$ under the differential. $\square$

8.17. Theorem. *Let G be a finite group, and k a field of characteristic $p > 0$. Then $\mathcal{Q}(G; k) = \mathcal{D}_b(kG)$. In particular, every kG -module admits a finite p -permutation resolution, and the canonical functor (see Recollection 1.11)*

$$\frac{K_b(\text{perm}(G; k)^{\natural})}{K_{b,ac}(\text{perm}(G; k)^{\natural})} \xrightarrow{\sim} \mathcal{D}_b(kG)$$

is an equivalence, i.e. the left-hand quotient is already idempotent-complete.

Proof. By the argument in Remark 8.9, all statements will follow once we show that $K_0^{\mathcal{Q}}(G; k) = G_0(kG)$. First, we claim that the subgroup $K_0^{\mathcal{Q}}(G; k) \subseteq G_0(kG)$ is an ideal. Indeed, by Proposition 8.10, it suffices to show that $K_0^{\mathcal{Q}}(G; k)$ is closed under multiplying by the class of a p -permutation module, which is straightforward.

We are therefore reduced to show that $1 = [k]$ belongs to $K_0^{\mathcal{Q}}(G; k)$. For this consider the Koszul complex $\text{Kos}(G; k)$ of Recollection 8.15, and set $K := \text{Kos}(G; k)_{\geq 1}[-1]$. By Lemma 8.16, K is a resolution of k by kG -modules. In degree 0 we have $K_0 = kG$ which is free. In degree $s + 1 > 0$ we have $K_{s+1} = \Lambda_k^s(kG)$ which is p -permutation by Lemma 8.13. This is enough by Proposition 2.22. $\square$

8.18. Corollary. *Let G be a p -group, where $p = \text{char}(k) > 0$. Then every finitely generated kG -module has a finite resolution by permutation kG -modules.* $\square$

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